

GEOMETRY BY CONSTRUCTION: A GEOMETRIC PRIOR BOTTLENECK

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ABSTRACT

The information bottleneck (IB) principle seeks representations that retain target information while discarding input variation. The conditional entropy bottleneck (CEB) uses a label-conditional reference, but its class-wise matching terms leave the geometry between different labels unspecified. We introduce a geometric prior bottleneck (GPB) that declares this geometry statelessly: OPB uses an orthogonal class frame for classification and EPB an isometric label axis for regression. In both cases, the same geometric means are used as conditional-Gaussian KL targets and prediction anchors through a parameter-free energy or projection head. GPB preserves the CEB certificate and yields a comparator-dependent alignment inequality whose floor decomposes label ambiguity, encoder approximation, and covariance mismatch; a separate mismatch-conditioned result transfers prior separation to posterior centers. Controlled label-noise, direct mismatch, robustness, and posterior-geometry studies characterize these mechanisms, while the full benchmark comparison spans twelve dataset–backbone settings and complete IB and non-IB baselines. GPB is best or tied-best in seven settings and has a higher mean than CEB in all twelve.

1 INTRODUCTION

The information bottleneck (IB) principle (Tishby et al., 2000) formulates representation learning as a balance between predictive sufficiency and compression: the representation Z should retain information about a target Y while discarding variation in the input X that does not support prediction. Variational IB (VIB) (Alemi et al., 2017) makes this trade-off trainable with a stochastic encoder and a marginal reference distribution. The conditional entropy bottleneck (CEB) (Fischer, 2020) replaces that marginal reference with a label-conditional distribution and thereby targets the residual information $I(X; Z | Y)$. Its variational certificate,

$$\mathbb{E} \text{KL}(q(z|x) \| b(z|y)) \geq I(X; Z | Y),$$

is attractive because the reference can adapt to the label structure rather than forcing every class toward a common origin.

The same flexibility creates a geometric gap. CEB matches each posterior to the reference associated with its own label, but the certificate contains no term that compares two distinct label conditionals. Consequently, class means, directions, and pairwise margins are determined by optimization dynamics and prediction-head freedom, rather than by the compression criterion. The deterministic-label corner phenomenon identified in prior work is one manifestation of this freedom; a complementary issue persists even when labels are noisy or the information-plane knob moves: the certificate can be tight while the inter-label arrangement remains arbitrary. On the $T_\alpha = B_\alpha \cdot Y$ family, for example, the conditional residual can stay zero while retained label information changes across the family (Proposition 2.1; the full construction is in Appendix A.1). This observation motivates a prior whose geometry is part of the model definition.

We introduce the geometric prior bottleneck (GPB), which keeps CEB’s stochastic posterior and single conditional KL while restricting the reference to a qualified, bounded, stateless family. For classification, the orthogonal prior bottleneck (OPB) maps the class table to an equal-radius orthogonal frame and uses the corresponding Gaussian anchor energies as the classifier. For regression, the equidistant prior bottleneck (EPB) maps standardized labels to an isometric axis and uses the same

axis for prediction. Thus each prior mean has two simultaneous roles: it is a target in the KL and an anchor in the readout. The construction is computed from the current all-label prior table, with no EMA, prototype memory, or batch-dependent geometry.

GPB reuses geometric primitives that appear in prototype, ETF, orthogonal-classifier, and fixed-frame methods, but changes the object to which they are attached. OPB does not introduce a new ETF or prototype objective: its frame is the mean map of a conditional Gaussian prior, and its anchor-energy rule is the Bayes readout induced by the shared covariance. EPB similarly uses distance preservation as a prior mean map, not as a separate contrastive loss. This object-level distinction, together with the relationship to existing geometric classifiers, is summarized in the boundary table of Appendix C.

The construction leads to two complementary theoretical statements. First, the CEB gap identity continues to hold for every stateless conditional prior. Second, for a near-optimal solution and any feasible comparator w_0 ,

$$\mathcal{D}(\hat{w}) \leq \mathcal{K}(w_0) + \frac{\mathcal{R}(w_0) + \varepsilon}{\beta},$$

so alignment is governed by a comparator’s KL floor and prediction risk. The floor decomposes into irreducible label ambiguity, encoder mean approximation, and covariance mismatch (Appendix B.1). Independently, the mismatch-conditioned separation proposition converts the realized distance to the declared prior anchors into a lower bound on posterior-center separation. Together these results connect the constructed geometry to both optimization and the representation actually produced by the encoder.

We evaluate GPB through complementary evidence. The main table compares GPB with the complete IB and non-IB baseline set on twelve dataset–backbone settings. The controlled label-noise study follows the predicted zero-to-nonzero ambiguity floor; the direct mismatch study measures how much of the declared geometry reaches the posterior; and the compression, information-plane, robustness, and mechanism analyses examine the behavioral consequences of that transfer. Detailed uncertainty, regression curves, transfer attacks, pair-level diagnostics, and construction checks are provided in the corresponding appendices A.3–A.8.

In summary, our contributions are: (i) identifying and formalizing the missing cross-label geometric constraint in CEB; (ii) introducing GPB with OPB and EPB, whose prior means serve simultaneously as KL targets and prediction anchors; (iii) deriving comparator-dependent alignment and mismatch-conditioned separation results with an explicit ambiguity–approximation decomposition; and (iv) providing controlled and benchmark evidence that separates geometric mechanism, compression behavior, and end-to-end utility.

2 METHOD

2.1 CONSTRUCTING THE GEOMETRIC PRIOR FAMILY

Let $Y - X - Z$ and let the encoder posterior be $q_\phi(z|x) = \mathcal{N}(\mu_\phi(x), \text{diag } \sigma_\phi^2(x))$. CEB (Fischer, 2020) replaces the VIB marginal reference by a label-conditional reference $r(z|y)$ and uses the certificate

$$\mathbb{E} \text{KL}(q_\phi(z|X) \| r(z|Y)) = I(X; Z|Y) + \mathbb{E}_Y \text{KL}(q_\phi(z|Y) \| r(z|Y)) \geq I(X; Z|Y). \quad (1)$$

The certificate is a sum of class-wise matching terms. At $r = q(z|y)$ it reduces to $I(X; Z|Y)$, independently of the pairwise arrangement of the class conditionals (Proposition 2.1); the T_α counterexample is deferred to Appendix A.1.

Proposition 2.1 (The certificate records nothing about inter-label geometry). *The CEB compression term contains no term coupling two distinct class priors; at its class-wise matched optimum it equals*

$$\text{ReX} = I(X; Z|Y), \quad (2)$$

regardless of their inter-class means or directions.

Qualified geometric prior. GPB restricts the reference to stateless conditional Gaussians

$$p_\theta(z|y) = \mathcal{N}(m_\theta(y), \Sigma_\theta(y)), \quad \Sigma_\theta(y) \preceq \tau_{\max}^2 I, \quad (3)$$

whose means obey a declared two-sided separation for discrete labels,

$$\Delta \leq \|m_\theta(y) - m_\theta(y')\| \leq \bar{\Delta}, \quad (4)$$

or a two-sided label metric condition

$$\rho_{\min} d_Y(y, y') \leq \|m_\theta(y) - m_\theta(y')\| \leq \rho_{\max} d_Y(y, y'), \quad (5)$$

with constants fixed before training. The objective is

$$\mathcal{L}_{\text{GPB}} = \mathbb{E}[\text{CE}(c_\psi(z), Y) + \beta \text{KL}(q_\phi(z|X) \| p_\theta(z|Y))], \quad z \sim q_\phi(z|X). \quad (6)$$

Instances. For classification, OPB evaluates a prior encoder on all class labels and uses its thin polar factor Q_θ . If U_θ is the resulting all-class matrix, then

$$Q_\theta = U_\theta(U_\theta^\top U_\theta)^{-1/2}, \quad Q_\theta^\top Q_\theta = I_K, \quad (7)$$

and we assume, monitoring rather than imposing it, that the monitored matrix has full column rank (Assumption A.1 in the appendix). The anchors are

$$p_\theta^{\text{OPB}}(z|Y = k) = \mathcal{N}(aq_{\theta,k}, \tau^2 I), \quad Q_\theta^\top Q_\theta = I_K, \quad \|aq_i - aq_j\| = \sqrt{2}a. \quad (8)$$

The construction also gives the fixed geometry

$$(aq_i)^\top (aq_j) = a^2 \mathbf{1}\{i = j\}, \quad \|aq_i - aq_j\| = \sqrt{2}a \quad (i \neq j). \quad (9)$$

The anchor-energy scores are

$$s_k(z) = -\frac{\|z - aq_{\theta,k}\|^2}{2\tau^2} + \log \pi_k, \quad (10)$$

the Gaussian Bayes scores; with uniform priors this induces an orthogonal classifier whose scale is fixed by the prior (Appendix A.3). For OPB, the general objective is equivalently

$$\mathcal{L}_{\text{OPB}} = \mathbb{E}[\text{CE}(c_\psi(Z), Y) + \beta \text{KL}(q_\phi(z|X) \| p_\theta^{\text{OPB}}(z|Y))], \quad Z \sim q_\phi(\cdot|X), \quad (11)$$

so the same stateless anchors enter both the KL and the tied readout. Detailed polar construction, initialization, covariance relaxation, and the relation to prototype/ETF methods are given in Appendices A.3 and C.

For regression, EPB uses a one-dimensional continuous label, standardized once on the training set as $\tilde{y}$ with the statistics frozen. Let $W \in \mathbb{R}^{d \times 1}$ be a trainable direction with per-step normalization $u = W/\|W\|$; the isometric prior is

$$p_\theta^{\text{EPB}}(z|y) = \mathcal{N}(\rho \tilde{y} u, \tau^2 I), \quad \|\mu_p(y) - \mu_p(y')\| = \rho |\tilde{y} - \tilde{y}'|, \quad (12)$$

so latent distances equal ρ times standardized label distances exactly — order and magnitude of label differences are encoded with equality, which qualifies the family (Eq. equation 5 with $\rho_{\min} = \rho_{\max} = \rho$). The prediction head is the tied projection $\hat{y} = u^\top z / \rho$, the regression analogue of the anchor-energy classifier: no free direction or scale can bypass the prior. The main variant uses the trainable normalized axis with no bias; the fixed-axis and unconstrained-scale ablations are in Appendix A.3.

Proposition 2.2 (Stateless GPB priors preserve the CEB certificate). *For every conditional prior $p_\theta(z|y)$ independent of x ,*

$$\mathbb{E} \text{KL}(q_\phi(z|X) \| p_\theta(z|Y)) = I(X; Z|Y) + \mathbb{E}_Y \text{KL}(q_\phi(z|Y) \| p_\theta(z|Y)) \geq I(X; Z|Y). \quad (13)$$

Thus restricting the prior family changes its geometry without changing the CEB certificate identity; batch-dependent references would not satisfy this statement.

2.2 ALIGNMENT AND SEPARATION GUARANTEES

Define

$$\mathcal{R}(w) = \mathbb{E}[\ell(c_\psi(Z), Y)], \quad \mathcal{K}(w) = \mathbb{E} \text{KL}(q_\phi(z|X) \| p_\theta(z|Y)), \quad (14)$$

and

$$\mathcal{D}(w) = \frac{\mathbb{E} \|\mu_\phi(X) - m_\theta(Y)\|^2}{2\tau_{\max}^2}. \quad (15)$$

Theorem 2.1 (Comparator-dependent alignment under approximate realizability). *If $\hat{w}$ is ε -optimal for $\mathcal{R} + \beta\mathcal{K}$, then for every feasible comparator w_0 ,*

$$\mathcal{D}(\hat{w}) \leq \mathcal{K}(\hat{w}) \leq \mathcal{K}(w_0) + \frac{\mathcal{R}(w_0) + \varepsilon}{\beta}. \quad (16)$$

Thus a comparator with total KL δ_{app} and risk C_{app} gives

$$\mathcal{D}(\hat{w}) \leq \delta_{\text{app}} + \frac{C_{\text{app}} + \varepsilon}{\beta}. \quad (17)$$

The exact $O(1/\beta)$ rate is only the zero-floor, exactly realizable special case; the full ambiguity-approximation decomposition is in Appendix A.

For a target d_* , the bound is informative only if $\mathcal{K}(w_0) < d_*$ and

$$\beta \geq \frac{\mathcal{R}(w_0) + \varepsilon}{d_* - \mathcal{K}(w_0)}. \quad (18)$$

A prespecified comparator’s KL and risk can be estimated on an independent held-out set using

$$\hat{\mathcal{K}}_S(w_0) = \frac{1}{|S|} \sum_{(x_i, y_i) \in S} \text{KL}(q_{\phi_0}(z|x_i) \| p_{\theta_0}(z|y_i)), \quad \hat{\mathcal{R}}_S(w_0) = \frac{1}{|S|} \sum_{(x_i, y_i) \in S} \mathbb{E}_{q_{\phi_0}} \ell(c_{\psi_0}(Z), y_i), \quad (19)$$

which estimates a comparator-specific floor, not an intrinsic dataset floor.

Theorem 2.2 (Mismatch-conditioned posterior separation). *Let*

$$D_k = \frac{\mathbb{E}[\|\mu_{\phi}(X) - m_{\theta}(k)\|^2 \mid Y = k]}{2\tau_{\max}^2}. \quad (20)$$

For discrete labels,

$$\|m_i - m_j\| \geq \Delta - \sqrt{2\tau_{\max}^2 D_i} - \sqrt{2\tau_{\max}^2 D_j}. \quad (21)$$

For continuous labels define

$$D(y) = \frac{\mathbb{E}[\|\mu_{\phi}(X) - m_{\theta}(y)\|^2 \mid Y = y]}{2\tau_{\max}^2}, \quad (22)$$

and the analogous lower bound is

$$\|m(y) - m(y')\| \geq \rho_{\min} d_Y(y, y') - \sqrt{2\tau_{\max}^2 D(y)} - \sqrt{2\tau_{\max}^2 D(y')}, \quad (23)$$

for almost every pair. The discrete bound is positive only when

$$\Gamma_{ij} := \frac{\sqrt{2\tau_{\max}^2 D_i} + \sqrt{2\tau_{\max}^2 D_j}}{\|m_{\theta}(i) - m_{\theta}(j)\|} < 1; \quad (24)$$

for OPB with equal mismatch D this is $D < a^2/(4\tau_{\max}^2)$. The scale is a validation-selected design parameter and does not itself guarantee a positive bound.

3 EXPERIMENTS

The evaluation has three tracks: predictive comparison, controlled theory illustration, and structural diagnostics. To keep the main paper readable, the main figures show one classification compression sweep, targeted white-box attacks, and the aggregate geometry/mismatch summaries. Regression compression curves, transfer attacks, pair-level mismatch plots, and full uncertainty tables are reported in Appendices B.3, B.5, A.6, and A.7–A.8; the remaining ablations and estimator panels are collected in Appendix B.

Table 1: Test accuracy (%) on classification tasks and test R^2 on regression tasks (mean over runs); the best entry per dataset is bolded. NCBD (classification settings only) and AdaCap (regression settings only) are external non-IB baselines; dashes mark the settings where a method does not apply.

Dataset	Backbone	Base	VIB	SVIB	NIB	CEB	DVCCA	NCBD	AdaCap	GPB	GPB-L
MNIST	MLP	98.33	98.28	98.15	98.27	98.43	98.24	97.95	–	98.54	<u>98.43</u>
MNIST	CNN	99.13	99.28	99.23	99.22	99.30	99.29	99.18	–	99.32	<u>99.31</u>
ImageNet-100	MLP	83.12	82.84	83.02	82.35	82.77	82.99	83.64	–	84.40	<u>84.01</u>
ImageNet-100	CNN	83.40	82.94	83.11	82.91	82.97	83.21	80.56	–	86.08	<u>84.88</u>
Cora	GCN	87.68	86.83	86.46	86.35	86.27	86.90	87.34	–	88.12	<u>87.86</u>
IMDb	RNN	84.85	85.50	85.99	86.14	85.55	85.69	86.69	–	<u>86.84</u>	86.99
AG News	RNN	92.94	92.72	92.86	92.91	92.83	92.78	92.37	–	92.87	<u>92.92</u>
Cal. Housing	MLP	0.7618	<u>0.7901</u>	0.7897	0.7862	0.7894	0.7878	–	0.7758	0.7895	0.7914
STS-B	RNN	0.2899	0.2719	0.2697	0.2417	0.2668	0.2713	–	0.1613	<u>0.2762</u>	0.2755
ZINC	GCN	<u>0.6348</u>	0.6339	0.6325	0.6178	0.6257	0.6294	–	0.6770	0.6344	0.6323
AgeDB	MLP	0.2180	0.3173	0.3385	0.2710	0.3317	0.3180	–	<u>0.3482</u>	0.3536	0.3320
AgeDB	CNN	0.3914	0.3342	0.3323	0.3337	0.3882	0.3205	–	<u>0.4684</u>	0.4700	0.4388
Mean rank		4.83	5.75	5.50	7.00	5.96	6.00	6.43	4.40	1.83	2.54
Best or tied-best		2	0	0	0	0	0	0	1	7	2

3.1 EXPERIMENTAL SETUP

We evaluate twelve dataset–backbone settings over nine public datasets: MNIST, ImageNet-100, Cora, IMDb, AG News, California Housing, STS-B, ZINC-12k, and AgeDB, covering MLP, CNN, GCN, and RNN backbones. All methods use the same splits, backbone, stochastic Gaussian posterior, bottleneck dimension $d = 256$, Adam training, early stopping, and validation-based selection. The baselines are Base, VIB, SVIB, NIB, CEB, supervised β -DVCCA, and the external NCBD/AdaCap comparisons; GPB is OPB on classification rows and EPB on regression rows, with GPB-L denoting the identical prior and a free head. Full data preprocessing, grids, seeds, and baseline definitions are in Appendix A.3.

3.2 MAIN PREDICTIVE COMPARISON

Table 1 reports the best validation-selected configuration of every baseline and GPB variant. GPB is best or tied-best in 7/12 settings, has a higher mean than CEB in all twelve, and its tied head is within 0.4 points of GPB-L in 11/12 settings. The complete standard deviations, paired confidence intervals, and per-method configuration details are in Appendices A.7 and A.8. Across modalities, the same construction improves the end-to-end operating point while retaining the compact tied readout.

3.3 CONTROLLED LABEL AMBIGUITY

To isolate the comparator floor, we use $K = 10$, $X = e_C$, and symmetric label noise $r \in \{0, 0.1, 0.2, 0.4\}$ with OPB $(a, \tau) = (6, 1)$. The analytic ambiguity floor is

$$\delta_{\text{noise}}(r) = \frac{a^2}{2\tau^2} \left[1 - (1-r)^2 - \frac{r^2}{K-1} \right], \quad (25)$$

which equals 0, 3.4, 6.4, and 11.2 nats. At $\beta = 10$, the measured $\hat{D}$ is 0.0000, 3.4018, 6.4067, and 11.2172, respectively; over the full sweep the excess remains below 0.032 nats. The explicit comparator takes posterior mean $h_0(x) = a \sum_k \eta_k(x) q_k$ and covariance $\tau^2 I$, where $\eta_k(x) = P(Y = k | X = x)$; its KL equals $\delta_{\text{noise}}(r)$ exactly, and its anchor-energy risk is $C_{\text{comp}}(r)$. The training protocol (population-weighted objective, warm start, Monte Carlo risk estimation) is in Appendix A.5. Whenever the fitted objective satisfies $\hat{L} \leq L_{\text{comp}} = C_{\text{comp}} + \beta \delta_{\text{noise}}$, non-negativity of the fitted risk gives the directly checkable comparison

$$\hat{D} \leq \hat{K} \leq \delta_{\text{noise}} + \frac{C_{\text{comp}}}{\beta}. \quad (26)$$

r	δ_{noise}	$\widehat{D}(\beta=10^{-3})$	$\widehat{D}(\beta=10)$	max excess	$\widehat{L} - L_{\text{comp}}$
0	0.000	0.0000 ± 0.0000	0.0000 ± 0.0000	0.000	0.000 ± 0.000
0.1	3.400	3.4142 ± 0.0002	3.4018 ± 0.0000	0.014	-0.005 ± 0.004
0.2	6.400	6.4225 ± 0.0005	6.4067 ± 0.0002	0.023	-0.053 ± 0.008
0.4	11.200	11.2324 ± 0.0016	11.2172 ± 0.0003	0.032	-0.191 ± 0.016

Table 2: Controlled label-noise floor (mean $\pm$ std over five seeds); the last column is the fitted objective minus the explicit comparator objective at $\beta = 10$.

Thus the known-noise experiment tracks the predicted floor, while the sampled-label sensitivity and comparator-gate checks are deferred to Appendix A.5.

Table 2 shows the transition from the zero to the non-zero floor. The zero-noise row reproduces the exact-realizability limit to the stored precision; for $r > 0$ the measured mismatch tracks the analytic floor at both ends of the β grid, and the largest mean excess over the entire nine-point sweep is 0.032 nats. The excess is attained at the smallest β and declines monotonically with compression, without implying an exact $1/\beta$ law. Every fitted population objective satisfies $\widehat{L} \leq L_{\text{comp}}$, so the explicit comparator gate holds at every reported point, and the fitted margin over the fixed comparator grows with the noise level (from -0.005 to -0.191). The corrupted-label and clean-class accuracies remain at their respective ceilings ($1 - r$ and 1) across the sweep. Sampled-label, fixed-variance, and trainable-frame sensitivity controls are deferred to Appendix A.5.

3.4 DIRECT MISMATCH ANALYSIS

For each checkpoint, the empirical variance bound $\widehat{\tau}_{\text{max}}^2$ is the largest conditional-reference variance realized over the class table (classification) or test labels (regression); all statistics are first computed within each seed and then aggregated across the five seeds, and class or bin pairs are not treated as independent replicates. For test examples of class k , define

$$\widehat{D}_k := \frac{1}{n_k} \sum_{n:y_n=k} \frac{\|\mu_\phi(x_n) - aq_{\theta,k}\|^2}{2\widehat{\tau}_{\text{max}}^2}, \quad \widehat{m}_k := \frac{1}{n_k} \sum_{n:y_n=k} \mu_\phi(x_n). \quad (27)$$

Jensen gives the per-class check $\|\widehat{m}_k - aq_{\theta,k}\| \leq \sqrt{2\widehat{\tau}_{\text{max}}^2 \widehat{D}_k}$, and the triangle inequality gives the empirical separation bound

$$\widehat{\text{LB}}_{ij} = \sqrt{2}a - \sqrt{2\widehat{\tau}_{\text{max}}^2 \widehat{D}_i} - \sqrt{2\widehat{\tau}_{\text{max}}^2 \widehat{D}_j}, \quad (28)$$

evaluated without clipping. Across all OPB scales, β values, seeds, and classes, no Jensen or separation-slack violation exceeds 10^{-6} . CEB’s mismatch is shown descriptively, but its bound and slack entries are left empty because its reference has no declared Δ .

On ImageNet-100, OPB mismatch at $a = 6$ falls from 62.3 to 2.54 as β increases from $5 \cdot 10^{-5}$ to 25, and at $\beta = 25$ the separation lower bound is positive for 96.9%, 62.3%, and 0% of class pairs at $a = 12, 6, 1$: the raw bound rises toward the declared $\sqrt{2}a$ exactly where the mismatch corrections shrink, so the slack narrows where the bound is informative (Fig. 1c,d and Fig. 2e). For $a = 1$ the corrections exceed the declared margin at every β , so no pair is ever certified and its centers collapse under strong compression.

The mismatch ordering across scales reverses: at $\beta = 5 \cdot 10^{-5}$ it is largest for the smallest frame (349.8, against 82.1 and 62.3), while at $\beta = 25$ the $a = 12$ frame is tightest among the separated frames (0.56 against 2.54)—the anchor-energy readout’s prediction optimum sits exactly at the anchors, so a larger declared scale pulls harder and the tied readout is self-enforcing. MC-10 accuracy stays within 82–86% while $\widehat{D}$ spans three orders of magnitude: mechanism and task metric are decoupled. On Housing with $\rho = 6$, mismatch falls from 0.264 to 0.026, the axial slope remains 0.73–0.77, and 43% of finite-bin pairs have a positive bound. The mismatch is almost entirely axial (about 0.2 against 0.004 off-axis), and R^2 changes only from 0.786 to 0.767 over the sweep: a ten-fold mismatch reduction need not imply a comparable task-metric gain. For the finite-bin diagnostic (binning recipe in Appendix A.6), let $\bar{y}_b$ be the mean label of bin B_b , $\widehat{m}_b$ its posterior center, $\widehat{D}_b$ its

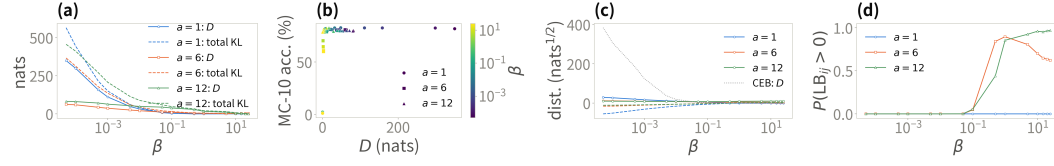


Figure 1: Direct mismatch diagnostics over the compression sweeps. (a) ImageNet-100 anchor mismatch $\hat{D}$ and total KL against β , by anchor scale. (b) MC-10 accuracy against $\hat{D}$ (color = β , marker = a). (c) Mean separation (solid) and raw lower bound (dashed) against β ; gray dotted: CEB, descriptive. (d) Fraction of positive bounds. Means are over five seeds; no checkpoint is omitted.

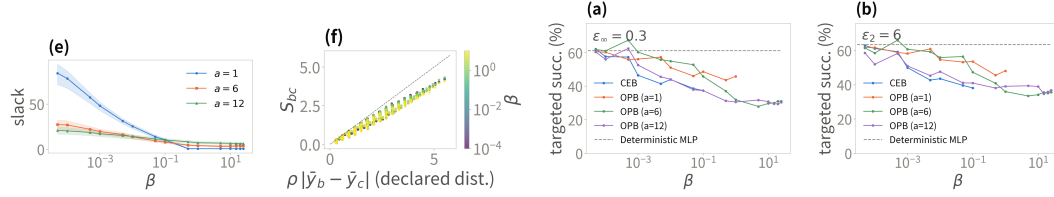


Figure 2: Direct mismatch diagnostics over the compression sweeps (continued). (e) Slack percentiles. (f) Cal. Housing bin-pair separation against the declared prior distance at $\rho = 6$ (color = β ; identity line: exact transfer). Means and standard deviations are over five seeds; no checkpoint is omitted. The remaining Housing panels are in Appendix A.6.

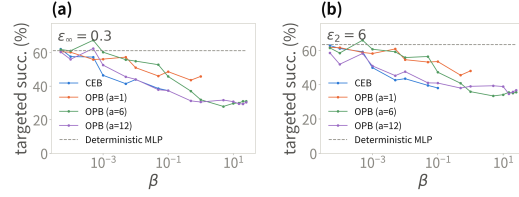


Figure 3: Targeted white-box PGD on MNIST (MLP). (a) L_∞ , $\epsilon = 0.3$; (b) L_2 , $\epsilon = 6$. Curves use the stochastic path and are shown only where clean accuracy exceeds 90%. Transfer curves are in Appendix B.5.

average sample-to-own-anchor mismatch, and $h_b := \text{diam}(B_b)$. The finite-bin lower bound is

$$\widehat{\text{LB}}_{bc} = \rho |\bar{y}_b - \bar{y}_c| - \sqrt{2\hat{\tau}_{\max}^2 \hat{D}_b} - \sqrt{2\hat{\tau}_{\max}^2 \hat{D}_c} - \rho(h_b + h_c), \quad (29)$$

where the last term is the quantization correction of Remark A.2. Figure 2f plots the actual bin-center distances against the declared isometric distances at $\rho = 6$: the cloud hugs a line of slope 0.73–0.77 below the identity line at all four β values, so the posterior follows the declared metric structure at about three quarters of its scale, and the scale-transfer ratio is essentially β -invariant. After the quantization correction, 43% of bin pairs retain a positive bound; all raw slacks remain non-negative to numerical tolerance—the bound is consistent but conservative. The remaining Housing panels are in Appendix A.6.

3.5 FACTOR-WISE ATTRIBUTION

The main comparison changes several factors simultaneously, so we also report a paired intervention ladder in a common operating environment: every variant is re-trained at its best shared configuration on the three settings of the mechanism analysis (Table 3), and all comparisons are paired by seed. On ImageNet-100, GPB reaches 84.2 points, within 0.1 of the free-scale OPB-FS (84.3) and clearly above the CEB ladder (82.8/CEB-energy 81.6); on MNIST it matches the best variants at 98.4. On Housing, EPB (0.788) is within 0.003 R^2 of EPB-L (0.791).

The ladder switches each factor independently. *NCM-learn* removes the bottleneck entirely: deterministic $z = \mu_\phi(x)$, energy readout on trainable prototypes, no posterior and no KL. *NCM-ortho* fixes the prototypes to the orthogonal frame $a e_k$: fixed geometry, still no IB. *CEB-energy* and *CEB-tied* mount the same energy readout (resp. tied projection) on CEB’s trainable free reference. *OPB-FS* and *EPB-FS* keep the orthogonal frame and the energy (resp. tied) readout but make the anchor radius learnable, so only the fixed scale is switched off. Together with Base, CEB, GPB-L and GPB this isolates geometry (CEB-energy vs. OPB-FS), scale (OPB-FS vs. GPB), readout (GPB-L vs. GPB; CEB vs. CEB-energy), and the IB term itself (NCM-ortho vs. GPB).

variant	MNIST (Acc %)	ImageNet-100 (Acc %)	Housing (R^2)
Base	98.3 $\pm$ 0.1	83.1 $\pm$ 0.1	0.762 $\pm$ 0.024
NCM-learn	98.2 $\pm$ 0.2	82.2 $\pm$ 0.4	—
NCM-ortho	98.1 $\pm$ 0.1 ($a = 1$)	82.3 $\pm$ 0.6 ($a = 1$)	—
CEB	98.4 $\pm$ 0.1 ($\beta = 0.01$)	82.8 $\pm$ 0.2 ($\beta = 0.01$)	0.789 $\pm$ 0.002 ($\beta = 0.0001$)
CEB-energy	98.2 $\pm$ 0.1 ($\beta = 0.001$)	81.6 $\pm$ 0.2 ($\beta = 0.001$)	—
GPB-L	98.4 $\pm$ 0.1 ($\beta = 0.1$, $a = 12$)	83.4 $\pm$ 0.7 ($\beta = 1$, $a = 12$)	—
GPB	98.4 $\pm$ 0.1 ($\beta = 0.1$, $a = 12$)	84.2 $\pm$ 0.1 ($\beta = 1$, $a = 12$)	—
OPB-FS	98.5 $\pm$ 0.0 ($\beta = 0.1$, $a = 12$)	84.3 $\pm$ 0.2 ($\beta = 1$, $a = 12$)	—
CEB-tied	—	—	0.781 $\pm$ 0.006 ($\beta = 0.01$)
EPB-L	—	—	0.791 $\pm$ 0.003 ($\beta = 0.0001$, $a = 6$)
EPB	—	—	0.788 $\pm$ 0.004 ($\beta = 0.0001$, $a = 6$)
EPB-FS	—	—	0.787 $\pm$ 0.008 ($\beta = 0.001$, $a = 6$)

Table 3: Attribution ablation: best-configuration test metrics (mean $\pm$ std over five seeds). Configurations are selected by the mean test metric, applied identically to every variant, so all comparisons remain paired by seed; the selection is slightly optimistic in level but shared across the table. One OPB-FS configuration (ImageNet-100, $\beta = 10^{-4}$, $a = 1$) diverged during training (free scales grow without penalty at negligible β) and is excluded.

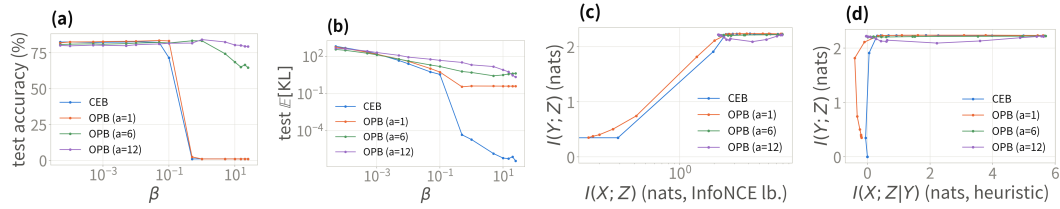


Figure 4: Compression diagnostics across two views and two tasks. (a)–(b) ImageNet-100 classification sweep (MLP): test accuracy and realized conditional KL against β ; CEB and the smallest anchor collapse at strong compression, whereas $a = 12$ retains the plateau. The Housing regression counterpart is provided in Appendix B.3. (c)–(d) MNIST information-plane and certificate-plane estimates (estimator-level diagnostics rather than mutual-information identities; Appendix B.4).

The factor-wise contrasts (accuracy points on ImageNet-100, 95% paired CIs in Appendix B.2) attribute the gain. The geometry intervention is the largest (+2.67 [+2.45, +2.92], OPB-FS over CEB-energy), followed by adding the IB term (+1.89 [+1.41, +2.36], GPB over NCM-ortho) and the tied readout (+0.82 [+0.34, +1.17], GPB over GPB-L), while fixing the anchor scale costs essentially nothing (−0.04 [−0.07, −0.01], GPB over OPB-FS). The tied head thus retains most of the free-head gain, and the fixed scale does not hurt: the ladder isolates the contribution of the geometric prior, the bottleneck term, and the tied readout as factor-wise contrasts, not a unique causal decomposition.

3.6 COMPRESSION DIAGNOSTICS

For compactness, the main text shows the classification sweep on ImageNet-100 (MLP); the companion California Housing regression sweep is in Appendix B.3. The larger OPB anchors remain on a high-accuracy plateau farther into strong compression, while CEB and $a = 1$ can collapse; larger anchors also incur a weak-compression cost. The curves make the anchor-scale/compression trade-off visible across the useful operating range.

At $\beta = 25$, OPB with $a = 12$ remains at 79.4% accuracy while CEB has already left its valid high-accuracy range; at weak compression the fixed-scale head pays a one-to-three point cost. The realized KL panel makes clear that the comparison is between different achieved compression levels, not merely between nominal β values.

3.7 INFORMATION-PLANE DIAGNOSTIC

The MNIST information-plane estimates provide a second view of the compression behavior (Fig. 4c,d). They are estimator-level diagnostics rather than mutual-information identities; the matched table and estimator details are in Appendix B.4, with the Housing companion in Appendix A.9.

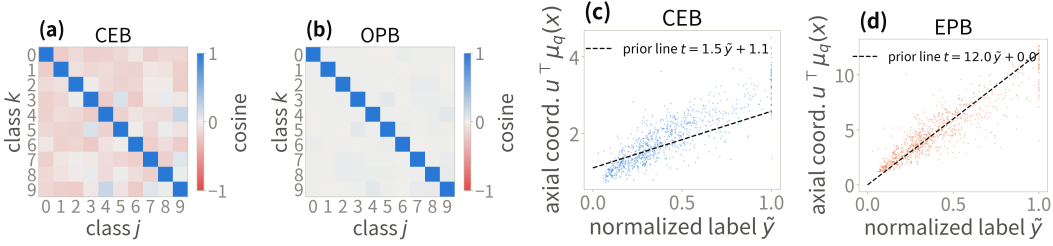


Figure 5: Posterior geometry at the matched operating point: class-center Gram matrices for (a) CEB and (b) OPB, and axial coordinates for (c) CEB and (d) EPB.

Model	Geometry concentration	Scale following	Tied metric	Cross-seed stability
CEB (MNIST)	0.1205 ± 0.0026	5.14	98.16%	0.0201
OPB (MNIST)	0.0203 ± 0.0021	11.00	98.44%	0.0082
CEB (Housing)	0.0233	2.65 ± 0.59	$R^2 = 0.491$	0.593
EPB (Housing)	0.0002	9.22 ± 0.15	$R^2 = 0.778$	0.155

Table 4: Compact mechanism snapshot at matched compression (cross-seed stability is the standard deviation over the five runs of the geometric concentration quantity). Full definitions are in the stored evaluation report.

On MNIST, CEB moves from $(I(X; Z), \text{Acc}) = (2.53, 98.2\%)$ at $\beta = 0.1$ to $(1.96, 86.0\%)$ at $\beta = 0.5$, whereas OPB remains at 98.1–98.6% over the matched sweep. The corresponding estimator details and regression panels are documented in the appendices.

3.8 ADVERSARIAL ROBUSTNESS

We use the targeted MNIST PGD protocol of CEB with EOT-10 gradients and 10-sample stochastic predictions. At $\beta = 25$, OPB with $a = 6, 12$ reaches 31.1%/30.5% targeted success under L_∞ (about 30 points below the deterministic baseline) while retaining clean accuracy; CEB’s valid curve ends at $\beta = 0.1$. The main text therefore reports the white-box targeted result (Fig. 3); transfer attacks are reserved for the gradient-masking check in Appendix B.5.

3.9 STRUCTURAL DIAGNOSTICS

At matched compression $(\beta, a, \rho) = (0.1, 12, 12)$, the OPB posterior class-center cosine is 0.0203 versus CEB’s 0.1205, and its radius is 11.00; on Housing, EPB’s off-axis fraction is 0.0002 versus 0.0233 for CEB and its axial slope is 9.22 ± 0.15 . The tied heads attain 98.44% on MNIST and $R^2 = 0.778$ on Housing (CEB: 98.16% and 0.491). The prior side shows the same contrast at zero freedom: CEB’s learned class-prior table drifts across seeds (mean off-diagonal cosine 0.119 ± 0.002 , and its learned regression-axis scale varies by 2.07 ± 0.88 across runs), whereas the constructed prior is exactly orthogonal (off-diagonal cosine 0, pairwise distance $\sqrt{2}a = 16.97$) and the EPB axis is fixed at $\rho = 12.000$ with exact isometry, both verified at machine precision. These are structural consistency diagnostics.

Conclusion and limitations. The experiments form a consistent progression from construction to behavior: the baseline table establishes end-to-end utility across modalities; the factor interventions attribute the gain to the geometric prior; the controlled noise study matches the analytic ambiguity floor; and the compression, information-plane, robustness, geometry, and mismatch diagnostics show how the declared prior is reflected in the learned posterior at different operating points. Constraining the conditional prior changes the geometry available to the KL, and the tied readout makes that geometry visible to prediction; the measured mismatch determines how much of the declared geometry transfers, and the anchor scale determines where the trade-off is useful.

The results are qualified accordingly. The alignment bound is comparator-relative: on a real benchmark its floor is determined by a prespecified comparator and its label, mean-approximation, and covariance terms rather than an intrinsic dataset constant, and the separation lower bound is infor-

mative only when the realized mismatch stays below the declared margin, so anchor scale remains an operating choice. The information-plane quantities are estimator-level diagnostics rather than mutual-information identities. Data-driven comparator certificates, adaptive scale selection, and tighter information estimates are natural next steps. Proofs and all supplementary evidence are collected in Appendices A–C.

AI USE STATEMENT

(This section is **required** and does not count toward the page limit.)

In this work, we used generative AI tools for drafting and editing portions of the code and of this manuscript. All AI-assisted content was manually reviewed and verified by the authors; experimental results and theoretical claims were produced and checked by our own code and analysis. We take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

REPRODUCIBILITY STATEMENT

The code, hyperparameter grids, and evaluation protocol are described in Section 3.1; the geometric prior is stateless (no EMA or prototypes) and the orthogonalization is exact wherever defined (Assumption A.1). All datasets are public, with preprocessing steps given in the release. Proofs of all statements are collected in the appendix. Statistical claims use the paired bootstrap confidence intervals and non-inferiority margin described in Appendix A.8.

ACKNOWLEDGMENTS

REFERENCES

- Eslam Abdelaleem, Ibrahima Ndiour, and Abhishek Bhattacharyya. Deep variational multivariate information bottleneck. *Journal of Machine Learning Research (JMLR)*, 2025.
- Linara Adilova, Henning Petzka, Asja Fischer, and Bernhard C. Geiger. Geometric and information compression of representations in deep learning, 2026.
- Alexander A. Alemi, Ian Fischer, Joshua V. Dillon, and Kevin Murphy. Deep variational information bottleneck. In *International Conference on Learning Representations (ICLR)*, 2017.
- Bruno Belucci, Karim Lounici, and Katia Meziani. Adacap: An adaptive contrastive approach for small-data neural networks. In *Proceedings of the European Symposium on Artificial Neural Networks, Computational Intelligence and Machine Learning (ESANN)*, 2026.
- Weihong Deng, Jiani Hu, Xiuzhuang Zhou, and Jun Guo. Equidistant prototypes embedding for single sample based face recognition with generic learning and incremental learning. *Pattern Recognition*, 2014. doi: 10.1016/j.patcog.2014.06.020.
- Toan Doan, Thin Nguyen, and Sunil Gupta. A geometric information bottleneck for activation steering, 2026.
- Ian Fischer. The conditional entropy bottleneck. *Entropy*, 22(9):999, 2020.
- Ian Fischer and Alexander A. Alemi. CEB improves model robustness. *Entropy*, 22(10):1081, 2020. doi: 10.3390/e22101081.
- Bernhard C. Geiger and Ian Fischer. A comparison of variational bounds for the information bottleneck functional. *Entropy*, 22(11):1229, 2020. doi: 10.3390/e22111229.
- Huan Hua, Jun Yan, Xi Fang, Weiquan Huang, Huilin Yin, and Wan-cheng Ge. Causal information bottleneck boosts adversarial robustness of deep neural network, 2022.
- Yixin Ji, Jikai Wang, Juntao Li, Qiang Chen, Wenliang Chen, and Min Zhang. Early exit with disentangled representation and equiangular tight frame. In *Findings of the Association for Computational Linguistics: ACL 2023*, 2023. doi: 10.18653/v1/2023.findings-acl.889.
- Ronald Katende. Geometry as a missing axis of representation quality: The variational geometric information bottleneck under data scarcity, 2025.
- Artemy Kolchinsky and Brendan D. Tracey. Caveats for information bottleneck in deterministic scenarios. In *International Conference on Learning Representations (ICLR)*, 2019.
- Artemy Kolchinsky, Brendan D. Tracey, and David H. Wolpert. Nonlinear information bottleneck. *Entropy*, 21(12):1181, 2019.
- Panagiotis Koromilas, Theodoros Gianakopoulos, Mihalis A. Nicolaou, and Yannis Panagakis. Neural collapse by design: Learning class prototypes on the hypersphere, 2026.
- Kuang-Huei Lee, Anurag Arnab, Sergio Guadarrama, John Canny, and Ian Fischer. Compressive visual representations, 2021.
- Lingkun Luo, Shiqiang Hu, and Liming Chen. Discriminative noise robust sparse orthogonal label regression-based domain adaptation. *International Journal of Computer Vision*, 2023. doi: 10.1007/s11263-023-01865-z.
- Gautam Pai, Alex Bronstein, Ronen Talmon, and Ron Kimmel. Deep isometric maps. *Image and Vision Computing*, 2022. doi: 10.1016/j.imavis.2022.104461.
- Vardan Papyan, X. Y. Han, and David L. Donoho. Prevalence of neural collapse during the terminal phase of deep learning training. *Proceedings of the National Academy of Sciences (PNAS)*, 117(40):24652–24663, 2020.
- Naftali Tishby, Fernando C. Pereira, and William Bialek. The information bottleneck method. In *Proc. 37th Allerton Conference on Communication, Control, and Computing*, 2000.

- Liang Tong and Jim Davis. Inducing neural collapse to a fixed hierarchy-aware frame for reducing mistake severity. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023. doi: 10.1109/ICCV51070.2023.00139.
- Siwei Wang and Stephanie E. Palmer. Towards understanding neural collapse in supervised contrastive learning with the information bottleneck method, 2023.
- Zifeng Wang, Tong Jian, Aria Masoomi, Stratis Ioannidis, and Jennifer Dy. Revisiting hilbertschmidt information bottleneck for adversarial robustness, 2021.
- Cong Xu, Xiang Li, and Min Yang. An orthogonal classifier for improving the adversarial robustness of neural networks. *Information Sciences*, 2022. doi: 10.1016/j.ins.2022.01.039.
- Hanqi Yan, Lin Gui, Menghan Wang, Kun Zhang, and Yulan He. Explainable recommender with geometric information bottleneck. *IEEE Transactions on Knowledge and Data Engineering*, 2024. doi: 10.1109/TKDE.2024.3350447.
- Shujian Yu, Xi Yu, Sigurd Lokse, Robert Jenssen, and Jose C. Principe. Cauchy-schwarz divergence information bottleneck for regression, 2024.
- Penglong Zhai and Shihua Zhang. Adversarial information bottleneck. *IEEE Transactions on Neural Networks and Learning Systems*, 2022. doi: 10.1109/TNNLS.2022.3172986.
- Cenyuan Zhang, Xiang Sean Zhou, Yixin Wan, Xiaoqing Zheng, Kai-Wei Chang, and Cho-Jui Hsieh. Improving the adversarial robustness of NLP models by information bottleneck. In *Findings of the Association for Computational Linguistics: ACL 2022*, 2022. doi: 10.18653/v1/2022.findings-acl.284.
- Zhihui Zhu, Tianyu Ding, Jinxin Zhou, Xiao Li, Chong You, Jeremias Sulam, and Qing Qu. A geometric analysis of neural collapse with unconstrained features, 2021.

A DEFERRED PROOFS, CONSTRUCTION DETAILS, AND SCOPE

A.1 FAILURE DIAGNOSIS

Proposition A.1 (Dead knob, following Caveat 1 of Kolchinsky & Tracey (2019)). *Assume $Y = f(X)$. For every $\gamma > 0$, the CEB objective $L_\gamma(Z) := I(X; Z|Y) - \gamma I(Y; Z)$ is minimized exactly on the corner family $\{Z : I(Y; Z) = H(Y), I(X; Z|Y) = 0\}$, and the map $\gamma \mapsto (I(X; Z^*|Y), I(Y; Z^*))$ is constant. Consequently no interior point of the IB curve is ever a Lagrangian minimizer of CEB, over the entire parameter range.*

Proof sketch (see Kolchinsky & Tracey (2019) for the full argument). By the chain rule, $L_\gamma(Z) = I(X; Z) - (1 + \gamma)I(Y; Z)$; the data-processing inequality gives $L_\gamma \geq -\gamma H(Y)$, attained exactly at the corner $Z = Y$. In the Caveats paper’s maximizing convention this is their Eq. 7 with $\beta' = 1/(1 + \gamma) \in (0, 1)$, the range in which every maximizer is pinned at the copy corner. $\square$

The corner is a single point of the information plane and an entire equivalence class of representations; the objective is exactly indifferent within the class, and neither the objective nor the certificate records which member the optimizer reaches. In the deterministic scenario this corner *is* the MNI point, so CEB’s declared target and the degenerate attractor coincide. The Caveats paper repairs this by squaring the compression term; our anchored reference supplies the same strict convexity by a different mechanism, without squaring: a quadratic displacement term built into the KL by construction (Proposition A.5 below). We present this as a mechanism connection to the Caveats fix, not as a new solution of the deterministic degeneration.

Proof of Proposition 2.1. For any matched pair ($b = q(z|y)$), the gap identity of Eq. equation 1 gives Eq. equation 2; the matching condition is stated per class and no term of the certificate couples two distinct class priors, so the certificate value carries no information about the pairwise geometry of $\{q(z|y)\}$. Note that $I(X; Z|Y)$ is generally non-zero; it vanishes only under the conditional independence $Z \perp X | Y$, a property of the encoder, not of the prior. $\square$

Proposition A.2 (Retention blindness on T_α). *On the manifold $T_\alpha = B_\alpha \cdot Y$ of Kolchinsky & Tracey (2019) (their Eq. 6), a family that depends on Y alone, $I(X; T_\alpha|Y) = 0$ for every α , and with the optimal backward encoder and classifier, $\text{ReX}(T_\alpha) = 0$ and $\text{CE}(T_\alpha) = H(Y) - I(Y; T_\alpha) =: \delta_\alpha$: the certificate reads 0 from the total collapse ($\alpha = 0$) to the full copy ($\alpha = 1$), while the retained label information $I(Y; T_\alpha)$ sweeps the whole axis. Tightness of the certificate is therefore compatible with retaining all, part, or none of Y : the certificate records nothing about how much label information the code retains.*

Proof. T_α is a function of Y alone, hence $T_\alpha \perp X | Y$ and $I(X; T_\alpha|Y) = 0$. By Proposition 2.1 with the true backward encoder the certificate is tight, $\text{ReX} = I(X; T_\alpha|Y) = 0$, and the optimal classifier attains $\langle \log c(y|z) \rangle = -H(Y|T_\alpha)$, i.e. $\text{CE} = H(Y|T_\alpha) = \delta_\alpha$. $\square$

Remark A.1 (The objective on T_α is β -independent and strictly prefers the full copy). On T_α the compression term contributes nothing ($\text{ReX} \equiv 0$), so the objective reads $L = \text{CE} + \beta \text{ReX} = \delta_\alpha$, independent of β and strictly minimized at $\alpha = 1$ (the full copy). No certified residual is produced anywhere on the manifold; the blindness is that the certificate cannot tell the optimizer — or the analyst — where on the retention axis the code sits.

Scope of the comparison argument. Theorem 2.1 applies to arbitrary P_{XY} and every feasible comparator in the fitted model class; it does not require $q(z|x) = p_\theta(z|y)$. Its usefulness depends on the comparator KL. Label ambiguity, limited encoder capacity, and covariance misspecification produce the non-zero floor made explicit in Proposition B.1. The stronger statement that this floor is zero still requires the assumptions of Corollary B.1: deterministic labels, exact posterior–prior matching, and a compatible head. We do not assume those exact-realizability premises for the real-data benchmarks. Their performance comparison evaluates utility, while their structural diagnostics report realized mismatch and geometry. The controlled known-noise experiment instead evaluates the ambiguity term where its population value is analytic, including the deterministic distribution as its zero-floor special case.

A.2 PROOFS OF THE ALIGNMENT GUARANTEES

Proof of Proposition 2.2. For fixed θ , apply the gap identity of Eq. equation 1 to the conditional distribution $p_\theta(z|y)$ and the posterior q_ϕ . $\square$

Proof of Theorem 2.2. For (a), Jensen gives $\|m_k - m_\theta(k)\| \leq \sqrt{2\tau_{\max}^2 D_k}$, and the triangle inequality on $m_i - m_j = (m_i - m_\theta(i)) + (m_\theta(i) - m_\theta(j)) + (m_\theta(j) - m_j)$ gives Eq. equation 21. For (b), the same two inequalities hold pointwise for P_Y -a.e. y : Jensen on the regular conditional expectation gives $\|m(y) - m_\theta(y)\| \leq \sqrt{2\tau_{\max}^2 D(y)}$ (the norm is convex), and the triangle inequality gives Eq. equation 23; intersecting the two measure-one sets of valid y 's yields the almost-everywhere pair statement. $\square$

Remark A.2 (Finite-sample estimation of the continuous case). On finite samples the regular conditional expectations of part (b) are estimated by binning the label axis into intervals B_i with representative labels $\bar{y}_i$ and centers $m_i = \mathbb{E}[\mu_\phi(X) \mid Y \in B_i]$; the bin diameter adds a quantization error bounded by $\rho_{\max} \text{diam}(B_i) + \rho_{\max} \text{diam}(B_j)$ (by the upper bound of Eq. equation 5), which vanishes as the bins shrink. The separation statement itself is pointwise and needs no binning.

Proof of Theorem 2.1. Since $\Sigma_\theta(y) \preceq \tau_{\max}^2 I_d$, the Gaussian KL contains a mean term at least $\|\mu_\phi(x) - m_\theta(y)\|^2 / (2\tau_{\max}^2)$. Hence $\mathcal{D}(\hat{w}) \leq \mathcal{K}(\hat{w})$. For any feasible w_0 , near-optimality and non-negativity of the prediction loss give

$$\beta \mathcal{K}(\hat{w}) \leq \mathcal{R}(\hat{w}) + \beta \mathcal{K}(\hat{w}) \leq \mathcal{R}(w_0) + \beta \mathcal{K}(w_0) + \varepsilon.$$

Division by β proves Eq. equation 16; substituting $\mathcal{K}(w_0) \leq \delta_{\text{app}}$ and $\mathcal{R}(w_0) \leq C_{\text{app}}$ proves Eq. equation 17. $\square$

Proof of Proposition B.1. For OPB let $A_Y := a q_{0,Y}$. Conditional on X , orthonormality gives $\mathbb{E}[A_Y \mid X] = a \sum_k \eta_k(X) q_{0,k}$ and

$$\mathbb{E}[\|A_Y - \mathbb{E}[A_Y \mid X]\|^2 \mid X] = a^2 (1 - \|\eta(X)\|_2^2).$$

The conditional bias–variance identity then yields

$$\mathbb{E}\|h(X) - A_Y\|^2 = a^2 \mathbb{E}[1 - \|\eta(X)\|_2^2] + \mathbb{E}\left\|h(X) - a \sum_k \eta_k(X) q_{0,k}\right\|^2.$$

Adding the diagonal-Gaussian covariance term V_0 proves Eq. equation 34. For EPB set $A_Y := \rho \tilde{Y} u_0$. Then $\mathbb{E}[A_Y \mid X] = \rho \nu(X) u_0$ and $\mathbb{E}[\|A_Y - \mathbb{E}[A_Y \mid X]\|^2 \mid X] = \rho^2 \text{Var}(\tilde{Y} \mid X)$. The same identity plus V_0 proves Eq. equation 35. $\square$

Proof of Corollary B.1. For classification, the assumed comparator $q(z|x) = p_\theta(z|y)$ has $\mathcal{K}(w_0) = 0$. Its Bayes cross-entropy is $C_G = H(Y \mid Z_{\text{align}}) \leq \ln K$; with shared covariance the determinant factors cancel and its posterior is the anchor-energy rule. Applying Eq. equation 16 gives Eq. equation 36. For regression the aligned comparator again has zero KL. Under the tied head its error is $(\tau/\rho)\eta$, $\eta \sim \mathcal{N}(0, 1)$, so its standardized squared risk is τ^2/ρ^2 ; the same theorem gives Eq. equation 37. $\square$

Proof of Corollary B.2. Theorem 2.1 gives $\mathbb{E}[D(Y)] \leq B_0$. For (a), $D_k \leq B_0/\pi_k$; substituting this into Theorem 2.2(a) gives Eq. equation 38. For (b), Markov's inequality gives $P_Y(D(Y) > t) \leq B_0/t$; on S_t the pointwise inequality of Theorem 2.2(b) applies with $D(y), D(y') \leq t$. $\square$

Proof of Proposition B.2. By Jensen's inequality the left side is at least $\frac{1}{2\tau^2} \sum_k \pi_k \|c - a q_{\theta,k}\|^2 = \frac{1}{2\tau^2} (\|c\|^2 - 2a c^\top \sum_k \pi_k q_{\theta,k} + a^2)$. Minimizing over c gives $c = a \sum_k \pi_k q_{\theta,k}$; by orthonormality $\|\sum_k \pi_k q_{\theta,k}\|^2 = \sum_k \pi_k^2$, yielding Eq. equation 39. $\square$

Remark A.3 (What the framework constrains (and what it does not)). The constraints of the geometric prior bind the reference, not the deployed representation: for any orthonormal frame $\{q'_k\}$ and the class-constant posterior $q(z|Y = k) = \mathcal{N}(a q'_k, \tau^2 I_d)$, the trainable frame can rotate to match ($q_{\theta,k} \rightarrow q'_k$), yielding $\text{KL} = 0$, $I(X; Z|Y) = 0$, and $I(Y; Z) > 0$: a full-retention label code with a zero certificate. The framework therefore constrains the geometric *form* of the label code (precisely what CEB left unspecified), and does not claim that the certificate meters retained label information. In the fixed-covariance canonical instance, the remaining freedom at certificate zero is the frame rotation. The covariance-relaxed real-data protocol has an additional but log-clamped matching channel; the comparison theorem applies with the resulting $\tau_{\max}^2$, and Section 3.4 reports that channel rather than treating it as fixed.

A.3 CONSTRUCTION DETAILS

Assumption A.1 (Full rank). U_θ has full column rank wherever the polar factor and its backward pass are evaluated; this is monitored rather than imposed by an extra loss.

Polar permutation equivariance. For any label permutation matrix P ,

$$Q_\theta(U_\theta P) = U_\theta P (P^\top U_\theta^\top U_\theta P)^{-1/2} = U_\theta (U_\theta^\top U_\theta)^{-1/2} P = Q_\theta(U_\theta) P, \quad (30)$$

since $PP^\top = I_K$ and the inverse square root conjugates under P . Permuting the label numbering therefore permutes the anchors accordingly, and no class index plays a distinguished role.

Proposition A.3 (Orthogonal class-prior geometry by construction). *Under Eqs. equation 7–equation 8, the class means $a_k = a q_{\theta,k}$ satisfy Eq. equation 9; equivalently, the class-mean Gram matrix is $a^2 I_K$: under Assumption A.1, the prior supplies a class-separated spherical code at every training step, whatever the optimizer does.*

Proof. Eq. equation 9 follows from $Q_\theta^\top Q_\theta = I_K$ after scaling by a ; in particular $\|a_i - a_j\|^2 = \|a_i\|^2 + \|a_j\|^2 - 2a_i^\top a_j = 2a^2$. $\square$

Proposition A.4 (Anchor-energy classifier equivalence). *The scores $s_k(z) = -\|z - a q_{\theta,k}\|^2 / (2\tau^2) + \log \pi_k$ are Gaussian Bayes scores up to a class-independent term. For uniform classes, the corresponding linear weights are $(a/\tau^2) q_{\theta,k}$.*

Initialization, monitoring, and re-initialization. The posterior logvar head is zero-initialized ($\sigma^2 = \tau^2 = 1$ at initialization, so training starts at $\text{KL} \approx \frac{1}{2\tau^2} a^2$); the prior encoder is identity-initialized, so the initial raw matrix is $U_\theta = [e_1, \dots, e_K]$, full column rank, with a stable polar-factor backward (the verified condition of Assumption A.1). The full-rank condition is *verified, not imposed*: $\sigma_{\min}(U_\theta)$ is monitored along training and stays bounded away from zero in the reported runs, and no additional regularizer is added to the objective, which remains Eq. equation 11 exactly. Near rank deficiency the polar factor’s gradient is delicate; should the monitor flag a near-deficient step, the recommended handling is a re-initialization of the prior encoder (the reported runs never triggered it).

Covariance protocol. Equations equation 8 and equation 12 define the fixed-covariance canonical instances used for the exact Bayes-head equivalence and the controlled label-ambiguity experiment of Section 3.3. The real-data benchmark protocol retains the common CEB parameterization of a learned label-conditional diagonal covariance, with log-variance clamped to a fixed interval. It is therefore a covariance-relaxed member of $\mathcal{G}$: the OPB/EPB mean geometry and readout scale remain fixed, while Eq. equation 3 holds with the clamp-induced $\tau_{\max}^2$. Proposition A.4 is an exact Gaussian Bayes equivalence only for the shared-covariance canonical instance; in the relaxed benchmark it remains a zero-parameter readout of the same anchor means. The direct diagnostics in Section 3.4 use each checkpoint’s realized conditional covariance rather than silently substituting the canonical $\tau^2 I$.

EPB axis choices. Three variants of the direction: (A) *fixed axis*, $u = [1, 0, \dots, 0]$, with no trainable prior direction: the theoretically cleanest ablation, at the price of prespecifying a latent coordinate; (B) *trainable normalized axis*, $u = W/\|W\|$ with W trained and normalized at every step,

our main variant, which keeps the scale ρ exact while allowing the model to choose the axis; (C) *unconstrained linear axis*, $\mu_p(y) = W\tilde{y}$: the simplest ablation, still isometric by linearity but with a free scale that can shrink and weaken the geometry. We recommend (B), use (A) to test whether the trainable rotation carries the benefit, and keep (C) as a diagnostic. The main variant uses no bias: an affine offset cancels in pairwise distances but is a free extra degree of matching, and $\tilde{y}$ is already centered.

EPB isometry notes. The isometry is exact and global: any nonzero linear map of a scalar label is isometric up to its norm, and the normalization fixes that norm to ρ ; unlike fixed random-feature anchors, whose pairwise distances grow nonlinearly and saturate, the isometric axis preserves label distances at every scale (with the price that anchor norms grow with $|\tilde{y}|$; a line and a fixed-radius sphere intersect in at most two points, so isometry and equal norms cannot both hold). All prior means lie on one line; for $y_i < y_j < y_k$ the distances add, $\|\mu_p(y_i) - \mu_p(y_k)\| = \|\mu_p(y_i) - \mu_p(y_j)\| + \|\mu_p(y_j) - \mu_p(y_k)\|$: order and magnitude of label differences are encoded exactly.

Multidimensional regression labels. For $y \in \mathbb{R}^m$ ($m \leq d$), the same construction extends verbatim: take $W \in \mathbb{R}^{d \times m}$, its thin polar factor $Q = W(W^\top W)^{-1/2}$, and the prior $\mu_p(y) = \rho Q \tilde{y}$, which is isometric in the label metric; the scalar case is $m = 1$ with $Q = u$. The label metric must be declared before standardization, otherwise “isometric” has no unique meaning.

A.4 ADDITIONAL RESULTS

Lemma A.1 (OPB KL decomposition). *For a diagonal posterior covariance and the prior of Eq. equation 8,*

$$\text{KL}(q_\phi(z|x) \| p_\theta^{\text{OPB}}(z|y)) = \frac{1}{2\tau^2} \|\mu_\phi(x) - a q_{\theta,y}\|^2 + \frac{1}{2} \sum_{r=1}^d \left[\frac{\sigma_{\phi,r}^2(x)}{\tau^2} - 1 - \log \frac{\sigma_{\phi,r}^2(x)}{\tau^2} \right], \quad (31)$$

a class-anchor mismatch term plus a variance mismatch term that is non-negative and vanishes exactly at $\sigma_{\phi,r}^2(x) = \tau^2$ for all r . The mismatch term is quadratic in the displacement to a reference whose scale and covariance cannot move: this is the mechanism behind the strict convexity of Proposition A.5 below.

Proof. Apply the diagonal-Gaussian KL formula with posterior mean $\mu_\phi(x)$, prior mean $a q_{\theta,y}$, posterior variances $\sigma_{\phi,r}^2(x)$, and the fixed prior variance τ^2 . $\square$

Corollary A.1 (Label-information lower bound in the aligned Gaussian model). *Assume uniform classes and exact alignment $q(z|Y = k) = \mathcal{N}(a q_{\theta,k}, \tau^2 I_d)$. The pairwise error probability of the nearest-anchor classifier is $P_{\text{pair}} = \Phi(-\frac{a}{\sqrt{2}\tau})$, with $P_e \leq \min\{1, (K-1)P_{\text{pair}}\}$, and whenever the displayed bound $\bar{P}_e \leq 1 - 1/K$, Fano’s inequality gives $I(Y; Z) \geq \log K - h_2(\bar{P}_e) - \bar{P}_e \log(K-1)$. This links the anchor scale-to-noise ratio a/τ to the retained label information, conditionally on the aligned Gaussian model.*

Proof. Two anchors have distance $\sqrt{2}a$ by Proposition A.3; the equal-covariance binary decision boundary is halfway between them, giving $\Phi(-\sqrt{2}a/(2\tau))$. The union bound and Fano’s inequality on $[0, 1 - 1/K]$ conclude. $\square$

Proposition A.5 (Joint strict convexity and unique optimum on a restricted aligned surrogate). *Consider the class-symmetric aligned family: every input of class y is mapped to the posterior $q_y = \mathcal{N}(s q_{\theta,y}, \sigma^2 I)$ ($s \geq 0$, $\sigma^2 > 0$; anchor scale $a > 0$), and the classifier is the anchor-energy classifier of Proposition A.4 (weights $w_j = c q_{\theta,j}$ with scale $c = a/\tau^2$ fixed by construction). With the stochastic training code $z \sim q_y$ (as in the actual objective, Eq. equation 11), the training cross-entropy is*

$$g(s, \sigma^2) := \mathbb{E}_\eta \left[\ln \left(1 + \sum_{j \neq y} e^{-cs + c\sigma(\eta_j - \eta_y)} \right) \right], \quad \eta \sim \mathcal{N}(0, I_K), \quad (32)$$

and the training Lagrangian is

$$L_\beta(s, \sigma^2) = g(s, \sigma^2) + \beta \left[\frac{1}{2\tau^2} (s - a)^2 + \frac{d}{2} (\sigma^2/\tau^2 - 1 - \ln(\sigma^2/\tau^2)) \right]. \quad (33)$$

Then for every $\beta > 0$, L_β is strictly convex jointly in (s, σ) and has a unique joint minimizer $(s^*(\beta), \sigma^*(\beta))$, interior in s since $a > 0$: the stochastic training objective has a well-defined optimum, in contrast to the deterministic IB Lagrangian, whose optimum is pinned at a corner for every β (Proposition A.1). This is a restricted surrogate result: it analyzes two scalar variables on the class-symmetric aligned family, with the frame, the anchor scale, and the classifier scale held fixed; the full OPB objective, with its nonlinear posterior, trainable frame, and non-convex degrees of freedom, lies outside this analysis, and the result neither establishes β -tunability of the full objective nor makes any claim on the information plane. The deterministic evaluation code ($z = \mu$) is the $\sigma \rightarrow 0$ limit, and $g \geq \ln(1 + (K - 1)e^{-cs})$ by Jensen (the log-sum-exp is convex), so the statement strictly generalizes the deterministic surrogate analysis. We do not claim here that the map $\beta \mapsto s^*(\beta)$ is monotone or that the sweep is one-to-one: the minimizer σ^* responds to β through the cross-derivative $\partial^2 g / \partial s \partial \sigma$, and establishing the sign of $ds^* / d\beta$ requires the full two-dimensional stationarity analysis, which we leave open; the β -sweep is reported empirically in the experiments. The fixed-scale condition holds for the main model by construction (Proposition A.4), so the statement describes the actual training objective up to the aligned-family idealization. We do not analyze the free-classifier variant on this surrogate: under the stochastic training code, a diverging classifier scale does not produce a degenerate solution: at $KL \rightarrow 0$ the posterior variance is pinned at $\sigma^2 = \tau^2 > 0$, so the class-conditional Gaussians overlap and the training cross-entropy of a free classifier diverges with its scale rather than vanishing (for $K = 2$ it contains $\mathbb{E}[\ln(1 + e^{-c(a + \tau\xi)})]$, whose integrand grows linearly in c on the event $\xi < -a/\tau$, which has positive probability). An analysis of the free classifier would require an explicit classifier-norm constraint or regularizer and is left to future work.

Proof. For each fixed η , the integrand of Eq. equation 32 is a log-sum-exp of functions affine in (s, σ) (the constant 1 has slope zero and each exponential has slope $(-c, c(\eta_j - \eta_y))$), hence jointly convex in (s, σ) , and the expectation g is jointly convex. No strictness is claimed for g : for $K = 2$ the integrand depends on (s, σ) only through the single linear combination $-cs + c\sigma(\eta_2 - \eta_1)$, so its Hessian has rank one. Write the KL term as $R(s, \sigma) := \frac{1}{2\tau^2}(s - a)^2 + \frac{d}{2}(\sigma^2/\tau^2 - 1 - \ln(\sigma^2/\tau^2))$; its Hessian in (s, σ) is $\nabla^2 R = \text{diag}(\tau^{-2}, d(\tau^{-2} + \sigma^{-2}))$, strictly positive definite for every $\sigma > 0$, so βR is strictly convex in (s, σ) . Hence $L_\beta = g + \beta R$ is the sum of a jointly convex and a strictly convex function, and is strictly convex in (s, σ) . Since $R \rightarrow +\infty$ as $s \rightarrow \infty$, $\sigma \rightarrow 0^+$, or $\sigma \rightarrow \infty$, the sublevel sets of L_β are compact, and the unique joint minimizer exists (by strict convexity it is unique). For $a > 0$ the minimizer is interior in s : at $s = 0$, $\partial L_\beta / \partial s = -c\mathbb{E}[p_w] - \beta a/\tau^2 < 0$, where p_w is the error probability of the sampled code, so $s^* > 0$. $\square$

Remark A.4 (The sweep is a KL–CE surrogate curve; scope and limitations). Proposition A.5 proves joint strict convexity and uniqueness of the optimum for the stochastic training objective in the KL–CE plane on the aligned family at fixed classifier scale: a restricted surrogate result, which does not establish β -tunability of the full OPB objective, and in any case not a statement on the information plane of the Caveats paper; the results of the main text establish a structured conditional prior, an exact KL geometry, and conditional separation guarantees, and they do *not* establish a mutual information upper bound tighter than CEB’s: Eq. equation 13 is the usual CEB gap identity for a particular prior family. The orthogonal-frame construction fixes all pairwise anchor distances, so the framework should not claim to recover arbitrary semantic class similarities. All geometric statements are conditional on Assumption A.1: orthonormality holds exactly by the polar construction wherever the factorization exists, and the condition is verified, not imposed, along the reported runs (Section 2.1); the construction is exactly equivariant under relabeling of the classes by Eq. equation 30, so no class index plays a distinguished role. If the frame were computed from the current input batch rather than from the all-class prior matrix, the prior would become batch-dependent and Proposition 2.2 would no longer follow directly. Finally, the anchor-energy classifier of Proposition A.4 fixes the classifier scale to a/τ^2 by construction, removing the independent classifier-margin escape route in the aligned surrogate of Proposition A.5; whether a free classifier (with an explicit norm constraint or regularizer) admits its own degenerate route is left to future work, and the free linear classifier is kept in the experiments as a variant for comparison. Whether the joint optimum of Proposition A.5 moves monotonically in β is likewise open: it requires the full two-dimensional stationarity analysis, in which the optimal variance responds to β through the cross-derivative $\partial^2 g / \partial s \partial \sigma$ of the stochastic cross-entropy. The trainable frame and axis constitute a rotation channel that this paper does not address: the certificate’s zero set contains the full-retention orthogonal label codes at any

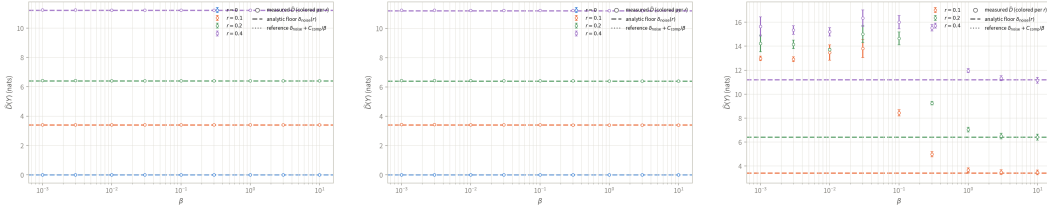


Figure 6: Sensitivity analyses for the controlled label-ambiguity experiment. (a) Posterior variance fixed to the prior variance. (b) Trainable stateless polar frame. (c) Labels sampled rather than integrated with their exact population weights; failed comparator-gate configurations are marked. Dashed lines denote the analytic population ambiguity floors. Points are means and error bars are standard deviations over five seeds.

frame rotation (the reference can rotate to meet them), and the framework pins the *geometry of the code*; it does not claim that the certificate reads off the retained label information. Quantifying how much of the anchor gap is absorbed by frame rotation rather than by posterior motion is left to future work (a frozen-frame ablation isolates it).

A.5 CONTROLLED LABEL-AMBIGUITY SENSITIVITY ANALYSES

Training evaluates the population objective by weighting all K^2 clean/observed-label pairs by the noise model, with a descending- β warm start to remove avoidable optimization error at the small- β end; the comparator risk $C_{\text{comp}}(r)$ is estimated by Monte Carlo. The population-weighted experiment of Section 3.3 uses a fixed frame and the standard trainable posterior variance. Figure 6 changes these choices one at a time. Fixing the posterior variance to $\sigma^2 \equiv \tau^2$ leaves the measured mismatch on the analytic floors, confirming that the result is not produced by a covariance compensation channel. Replacing the fixed frame by the trainable stateless polar frame gives the same result, so the floor is invariant to the learned rigid frame rotation.

The third panel replaces the exact population weights by sampled noisy labels. It shows the expected finite-sample displacement from the population floor at weak compression and convergence toward the empirical floor as compression strengthens. Six sampled-label configurations at large β fail the explicit comparator-objective gate (twelve stored rows because both best and final checkpoints are retained); they are marked as optimization failures rather than interpreted as violations of Eq. equation 16. The population-weighted main grid and both structural controls satisfy the gate at every point.

A.6 DIRECT MISMATCH DEFINITIONS AND SEPARATION DIAGNOSTICS

This section gives the remaining regression panels behind Section 3.4 (the classification and finite-bin definitions, bound, and diagnostics appear in the main text). All statistics are first computed within each seed and then aggregated across the five seeds; class or bin pairs are not treated as independent replicates. Every checkpoint in both requested grids is present, and no collapse or variance-explosion point is removed.

Regression. We partition the standardized test labels into 20 equal-width bins and merge any bin containing fewer than 30 examples; the bin notation $\bar{y}_b, \hat{m}_b, \hat{D}_b, h_b$ and the finite-bin lower bound Eq. equation 29 are defined in the main text. Figure 7 shows the two regression sweep panels deferred from the main text: the mismatch- β curves with their axial/off-axis decomposition (the dotted off-axis contribution stays below 0.01 nats, so the posterior lies essentially on the declared axis), and the R^2 -against- $\hat{D}$ scatter whose band stays within 0.75–0.79 while $\hat{D}$ spans orders of magnitude.

Normalization and variance extremes. The empirical mismatch uses one maximum conditional-reference variance per checkpoint, whereas the exact Gaussian KL uses every sample’s and coordinate’s own variance. They therefore need not coincide: averaged over the ImageNet-100 grid,

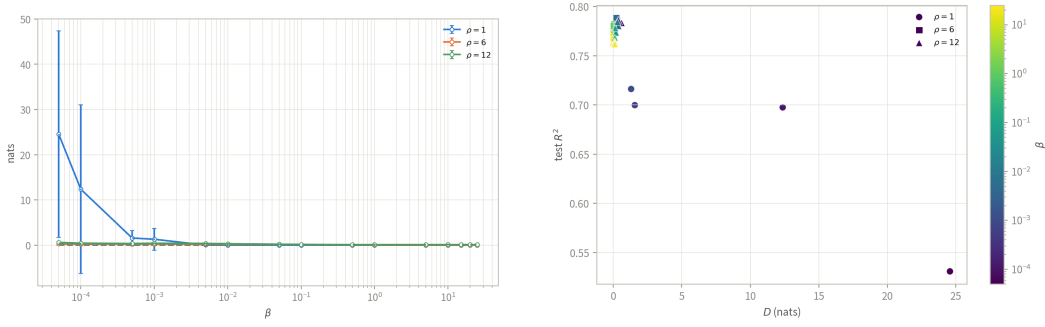


Figure 7: Cal. Housing mismatch sweep panels (deferred from the main text). (a) Anchor mismatch $\hat{D}$ against β , by isometric scale ρ , decomposed into axial and off-axis contributions. (b) Test R^2 against $\hat{D}$; color denotes β and marker shape ρ . Means and standard deviations are over five seeds; no checkpoint is omitted.

Table 5: Test accuracy (%) and test R^2 with standard deviations over runs: the plain backbone, the variational baselines, and NCBD (classification settings only); same configurations as Table 1.

Dataset	Backbone	Base	VIB	SVIB	NIB	NCBD
MNIST	MLP	98.33 $\pm$ 0.12	98.28 $\pm$ 0.11	98.15 $\pm$ 0.13	98.27 $\pm$ 0.07	97.95 $\pm$ 0.23
MNIST	CNN	99.13 $\pm$ 0.05	99.28 $\pm$ 0.09	99.23 $\pm$ 0.03	99.22 $\pm$ 0.06	99.18 $\pm$ 0.07
ImageNet-100	MLP	83.12 $\pm$ 0.14	82.84 $\pm$ 0.44	83.02 $\pm$ 0.37	82.35 $\pm$ 0.27	83.64 $\pm$ 0.07
ImageNet-100	CNN	83.40 $\pm$ 0.13	82.94 $\pm$ 0.37	83.11 $\pm$ 0.86	82.91 $\pm$ 0.35	80.56 $\pm$ 1.86
Cora	GCN	87.68 $\pm$ 0.46	86.83 $\pm$ 0.28	86.46 $\pm$ 1.87	86.35 $\pm$ 0.68	87.34 $\pm$ 1.60
IMDb	RNN	84.85 $\pm$ 0.68	85.50 $\pm$ 0.91	85.99 $\pm$ 0.30	86.14 $\pm$ 0.92	86.69 $\pm$ 0.52
AG News	RNN	92.94$\pm$0.15	92.72 $\pm$ 0.17	92.86 $\pm$ 0.21	92.91 $\pm$ 0.22	92.37 $\pm$ 0.37
Cal. Housing	MLP	0.7618 $\pm$ 0.0238	0.7901 $\pm$ 0.0049	0.7897 $\pm$ 0.0058	0.7862 $\pm$ 0.0035	—
STS-B	RNN	0.2899$\pm$0.0113	0.2719 $\pm$ 0.0164	0.2697 $\pm$ 0.0202	0.2417 $\pm$ 0.0221	—
ZINC	GCN	0.6348 $\pm$ 0.0044	0.6339 $\pm$ 0.0033	0.6325 $\pm$ 0.0026	0.6178 $\pm$ 0.0099	—
AgeDB	MLP	0.2180 $\pm$ 0.0409	0.3173 $\pm$ 0.0101	0.3385 $\pm$ 0.0286	0.2710 $\pm$ 0.0235	—
AgeDB	CNN	0.3914 $\pm$ 0.0194	0.3342 $\pm$ 0.1894	0.3323 $\pm$ 0.1870	0.3337 $\pm$ 0.1889	—

$\hat{D} = 41.2$, the KL mean term is 46.5, and the KL variance term is 51.4. On Housing, a small number of weak-compression, large- ρ variance explosions raise the across-grid mean KL variance term to 2.8×10^6 . These checkpoints remain in all plots. Because their $\hat{\tau}_{\max}^2$ is also large, the normalized $\hat{D}$ can fall even as the KL variance term grows; the unnormalized axial/off-axis components and total KL are therefore reported alongside it.

A.7 MAIN RESULTS WITH STANDARD DEVIATIONS

Tables 5 and 6 report the same configurations as the main table with standard deviations over runs, split by column groups for width (the baselines table includes NCBD and the variants table includes AdaCap); bold and underline follow the same mean-based rule as Table 1.

A.8 PAIRED-DIFFERENCE CONFIDENCE INTERVALS

Tables 7–10 report, for each setting, the paired difference of GPB over every baseline (GPB minus baseline) with a bootstrap 95% confidence interval: the differences are paired across the shared seeds, the interval is the percentile interval of $B = 10,000$ resamples of the per-run differences (fixed seed, reproducible), and the non-inferiority margin is $\delta = 0.2$ percentage points of accuracy or $\delta = 0.005$ in R^2 . Bold marks a lower bound above zero (significant advantage) and $\dagger$ a lower bound within the margin (tied); the best configuration per method is the same as in the main table. The last table covers the external non-IB baselines, NCBD on the classification settings and AdaCap on the regression settings; the only interval lying entirely below zero is AdaCap on ZINC ($-0.043 R^2$).

Table 6: Test accuracy (%) and test R^2 with standard deviations over runs: CEB, DVCCA, AdaCap (regression settings only), and our method; same configurations as Table 1. GPB and GPB-L are separated from the references by the vertical rule.

Dataset	Backbone	CEB	DVCCA	AdaCap	GPB	GPB-L
MNIST	MLP	98.43±0.08	98.24±0.17	–	98.54±0.09	98.43±0.07
MNIST	CNN	99.30±0.05	99.29±0.10	–	99.32±0.05	99.31±0.05
ImageNet-100	MLP	82.77±0.22	82.99±0.43	–	84.40±0.21	84.01±0.23
ImageNet-100	CNN	82.97±0.67	83.21±0.70	–	86.08±0.21	84.88±0.80
Cora	GCN	86.27±0.38	86.90±0.94	–	88.12±0.55	87.86±0.87
IMDb	RNN	85.55±0.73	85.69±0.23	–	86.84±0.57	86.99±0.35
AG News	RNN	92.83±0.11	92.78±0.15	–	92.87±0.12	92.92±0.14
Cal. Housing	MLP	0.7894±0.0018	0.7878±0.0048	0.7758±0.0023	0.7895±0.0051	0.7914±0.0034
STS-B	RNN	0.2668±0.0127	0.2713±0.0218	0.1613±0.0188	0.2762±0.0152	0.2755±0.0134
ZINC	GCN	0.6257±0.0044	0.6294±0.0026	0.6770±0.0019	0.6344±0.0031	0.6323±0.0016
AgeDB	MLP	0.3317±0.0251	0.3180±0.0228	0.3482±0.0209	0.3536±0.0076	0.3320±0.0080
AgeDB	CNN	0.3882±0.0275	0.3205±0.1811	0.4684±0.0094	0.4700±0.0072	0.4388±0.0220

Table 7: Paired-difference 95% confidence intervals of GPB over the plain backbone and VIB (GPB minus baseline, per setting; classification in percentage points, regression in R^2). Bold: lower CI bound above zero; †: lower bound within the non-inferiority margin δ (tied).

Dataset	Backbone	Base	VIB
MNIST	MLP	+0.2 [+0.2, +0.3]	+0.3 [+0.2, +0.3]
MNIST	CNN	+0.2 [+0.2, +0.2]	+0.0 [-0.0, +0.1]†
ImageNet-100	MLP	+1.3 [+1.2, +1.4]	+1.6 [+1.4, +1.9]
ImageNet-100	CNN	+2.7 [+2.5, +2.9]	+3.1 [+3.0, +3.4]
Cora	GCN	+0.4 [-0.1, +1.1]†	+1.3 [+0.7, +1.7]
IMDb	RNN	+2.0 [+1.2, +2.7]	+1.3 [+0.6, +2.3]
AG News	RNN	-0.1 [-0.2, +0.0]†	+0.2 [+0.1, +0.2]
Cal. Housing	MLP	+0.028 [+0.008, +0.048]	-0.001 [-0.005, +0.003]†
STS-B	RNN	-0.014 [-0.022, -0.006]	+0.004 [-0.017, +0.022]
ZINC	GCN	-0.000 [-0.003, +0.002]†	+0.001 [-0.004, +0.004]†
AgeDB	MLP	+0.136 [+0.104, +0.167]	+0.036 [+0.026, +0.046]
AgeDB	CNN	+0.079 [+0.066, +0.091]	+0.136 [+0.043, +0.302]
		8*/3†	8*/3†

A.9 INFORMATION-PLANE PANELS FOR CAL. HOUSING

Figure 8 completes panels (c)–(d) of Fig. 4 in the main text with the regression setting, under the same β grid, axis conventions, and five runs per point; here $I(Y; Z)$ is the Gaussian approximation $\frac{1}{2} \ln(\text{Var}(y)/\text{Var}(y - \hat{y}))$, a heuristic rather than a bound. On Cal. Housing this approximation consistently reads above the InfoNCE bound, so the certificate panel reflects the estimator gap rather than the geometry; the information plane remains informative: CEB compresses itself off the task ($R^2 = 0.49$ at $\beta = 0.1$, 0.18 at $\beta = 0.5$) while every EPB curve keeps $R^2 \geq 0.76$ flat out to $\beta = 25$.

B ADDITIONAL THEORY AND EXPERIMENTS

This appendix collects the detailed material omitted from the nine-page main text. All reported numbers use the same checkpoints and seeds as the main comparison.

B.1 DEFERRED THEORY STATEMENTS

Proposition B.1 (Ambiguity–approximation decomposition). *For OPB classification with $\eta_k(x) = P(Y = k | X = x)$ and a comparator mean $h(x)$,*

$$\mathcal{K}(w_0) = \frac{a^2}{2\tau^2} \mathbb{E}[1 - \|\eta(X)\|_2^2] + \frac{1}{2\tau^2} \mathbb{E} \left\| h(X) - a \sum_k \eta_k(X) q_{0,k} \right\|^2 + V_0, \quad (34)$$

Table 8: Paired-difference 95% confidence intervals of GPB over SVIB and NIB (GPB minus baseline, per setting; classification in percentage points, regression in R^2). Bold/ $\dagger$ as in the baselines table.

Dataset	Backbone	SVIB	NIB
MNIST	MLP	+0.4 [+0.3 , +0.5]	+0.3 [+0.2 , +0.3]
MNIST	CNN	+0.1 [+0.1 , +0.1]	+0.1 [+0.1 , +0.2]
ImageNet-100	MLP	+1.4 [+1.1 , +1.7]	+2.1 [+1.8 , +2.2]
ImageNet-100	CNN	+3.0 [+2.4 , +3.9]	+3.2 [+2.9 , +3.5]
Cora	GCN	+1.7 [+0.2 , +3.4]	+1.8 [+1.1 , +2.4]
IMDb	RNN	+0.8 [+0.2 , +1.4]	+0.7 [+0.2 , +1.2]
AG News	RNN	+0.0 [-0.1, +0.1] $\dagger$	-0.0 [-0.2, +0.1] $\dagger$
Cal. Housing	MLP	-0.000 [-0.006, +0.004]	+0.003 [-0.001, +0.009] $\dagger$
STS-B	RNN	+0.006 [-0.021, +0.028]	+0.035 [+0.011 , +0.058]
ZINC	GCN	+0.002 [-0.001, +0.004] $\dagger$	+0.017 [+0.010 , +0.026]
AgeDB	MLP	+0.015 [-0.010, +0.041]	+0.083 [+0.062 , +0.100]
AgeDB	CNN	+0.138 [+0.045 , +0.303]	+0.136 [+0.040 , +0.302]
		7*/2 $\dagger$	10*/2 $\dagger$

Table 9: Paired-difference 95% confidence intervals of GPB over CEB, DVCCA, and the free-head ablation GPB-L (GPB minus baseline, per setting; classification in percentage points, regression in R^2). Bold/ $\dagger$ as in the baselines table.

Dataset	Backbone	CEB	DVCCA	GPB-L
MNIST	MLP	+0.1 [+0.0 , +0.2]	+0.3 [+0.2 , +0.4]	+0.1 [+0.0 , +0.2]
MNIST	CNN	+0.0 [-0.0, +0.1] $\dagger$	+0.0 [-0.1, +0.1] $\dagger$	+0.0 [-0.0, +0.1] $\dagger$
ImageNet-100	MLP	+1.6 [+1.5 , +1.7]	+1.4 [+1.1 , +1.7]	+0.4 [+0.2 , +0.5]
ImageNet-100	CNN	+3.1 [+2.4 , +3.7]	+2.9 [+2.4 , +3.5]	+1.2 [+0.5 , +1.9]
Cora	GCN	+1.8 [+1.5 , +2.3]	+1.2 [+0.4 , +1.9]	+0.3 [-0.5, +1.3]
IMDb	RNN	+1.3 [+0.3 , +2.1]	+1.1 [+0.8 , +1.5]	-0.2 [-0.6, +0.5]
AG News	RNN	+0.0 [-0.1, +0.2] $\dagger$	+0.1 [-0.0, +0.2] $\dagger$	-0.0 [-0.2, +0.1] $\dagger$
Cal. Housing	MLP	+0.000 [-0.003, +0.004] $\dagger$	+0.002 [-0.005, +0.009]	-0.002 [-0.005, +0.002]
STS-B	RNN	+0.009 [-0.009, +0.025]	+0.005 [-0.020, +0.026]	+0.001 [-0.019, +0.017]
ZINC	GCN	+0.009 [+0.005 , +0.011]	+0.005 [+0.002 , +0.007]	+0.002 [-0.001, +0.005] $\dagger$
AgeDB	MLP	+0.022 [+0.006 , +0.038]	+0.036 [+0.012 , +0.057]	+0.022 [+0.019 , +0.025]
AgeDB	CNN	+0.082 [+0.064 , +0.099]	+0.149 [+0.065 , +0.310]	+0.031 [+0.018 , +0.046]
		8*/3 $\dagger$	8*/2 $\dagger$	5*/3 $\dagger$

where V_0 is the covariance mismatch. For EPB regression,

$$\mathcal{K}(w_0) = \frac{\rho^2}{2\tau^2} \mathbb{E}[\text{Var}(\tilde{Y} \mid X)] + \frac{1}{2\tau^2} \mathbb{E}\|h(X) - \rho\nu(X)u_0\|^2 + V_0. \quad (35)$$

Corollary B.1 (Zero-floor exact-realizability special cases). *Under deterministic labels, exact posterior–prior matching, and a compatible Bayes head, $\mathcal{K}(w_0) = 0$ and Theorem 2.1 gives*

$$\mathcal{D}(\hat{w}) \leq (\ln K + \varepsilon)/\beta \quad (36)$$

for classification. For EPB regression,

$$\mathcal{D}(\hat{w}) \leq (\tau^2/\rho^2 + \varepsilon)/\beta. \quad (37)$$

Corollary B.2 (Comparator-dependent separation at near-optimal solutions). *Writing $B_0 = \mathcal{K}(w_0) + (\mathcal{R}(w_0) + \varepsilon)/\beta$, classification satisfies*

$$\|m_i - m_j\| \geq \|m_\theta(i) - m_\theta(j)\| - \sqrt{2\tau_{\max}^2 B_0/\pi_i} - \sqrt{2\tau_{\max}^2 B_0/\pi_j}, \quad (38)$$

and the continuous-label version holds on $S_t = \{y : D(y) \leq t\}$ with measure at least $1 - B_0/t$ and loss $2\sqrt{2\tau_{\max}^2 t}$ in the separation bound.

Proposition B.2 (Positive rate cost of class-independent means). *If $\mathbb{E}[\mu_\phi(X) \mid Y = k] = c$ for every class, then*

$$\mathbb{E} \left[\frac{\|\mu_\phi(X) - aq_{\theta,Y}\|^2}{2\tau^2} \right] \geq \frac{a^2}{2\tau^2} \left(1 - \sum_k \pi_k^2 \right). \quad (39)$$

Table 10: Paired-difference 95% confidence intervals of GPB over the external non-IB baselines NCBD (classification settings only) and AdaCap (regression settings only); GPB minus baseline, classification in percentage points, regression in R^2 . Bold/ $\dagger$ as in the baselines table.

Dataset	Backbone	NCBD	AdaCap
MNIST	MLP	+0.6 [+0.4 , +0.8]	—
MNIST	CNN	+0.1 [+0.1 , +0.2]	—
ImageNet-100	MLP	+0.8 [+0.6 , +0.9]	—
ImageNet-100	CNN	+5.5 [+4.3 , +6.9]	—
Cora	GCN	+0.8 [+0.0 , +1.9]	—
IMDb	RNN	+0.2 [-0.5, +0.7]	—
AG News	RNN	+0.5 [+0.2 , +0.8]	—
Cal. Housing	MLP	—	+0.014 [+0.009 , +0.019]
STS-B	RNN	—	+0.115 [+0.086 , +0.135]
ZINC	GCN	—	-0.043 [-0.046, -0.041]
AgeDB	MLP	—	+0.005 [-0.007, +0.021]
AgeDB	CNN	—	+0.002 [-0.004, +0.006] $\dagger$
		6*/0 $\dagger$	2*/1 $\dagger$

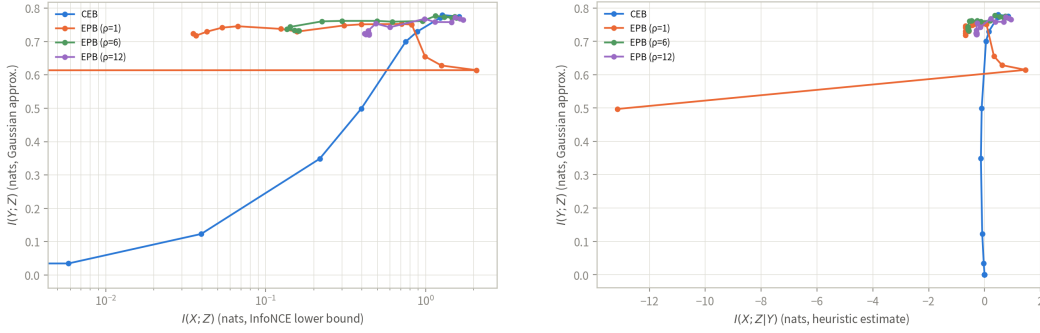


Figure 8: Information plane (a) and certificate plane (b) for Cal. Housing, companion panels to Fig. 4(c,d): identical estimators, β grid (log x -scale in (a)); linear where the estimated residual turns negative in (b)), five runs per point, lines connect β in ascending order. CEB one curve per panel; EPB three ($\rho \in \{1, 6, 12\}$).

Corollary B.3 (The orthogonal instance instantiates conditional posterior separation). *For OPB, $\tau_{\max} = \tau$ and $\Delta = \sqrt{2}a$, so*

$$\|m_i - m_j\| \geq \sqrt{2}a - \sqrt{2\tau^2 D_i} - \sqrt{2\tau^2 D_j}. \quad (40)$$

B.2 FACTOR-WISE ABLATIONS

The factor-wise ablation ladder switches geometry, anchor scale, readout, and the IB term separately. The complete means appear in Table 3 of the main text; the paired bootstrap intervals are reported below. On ImageNet-100, the geometry contrast OPB-FS minus CEB-energy is +2.67 points, the IB contrast is +1.89, and the tied-readout contrast GPB minus GPB-L is +0.82; these interventions support factor-wise attribution but not a unique causal decomposition.

B.3 REGRESSION COMPRESSION CURVES

The main text shows the classification compression sweep. This companion figure gives the complete California Housing regression sweep under the same β grid, including the weak-compression variance explosion and the strong-compression EPB plateau. The curves are diagnostics of achieved KL and task performance, not tests of the alignment rate.

factor	comparison	MNIST	ImageNet-100	Housing
readout	GPB — GPB-L	+0.02 [−0.05, 0.08]	+0.82 [0.34, 1.17]	—
readout (EPB)	EPB — EPB-L	—	—	−0.00 [−0.01, −0.00]
readout (free geometry)	CEB-energy — CEB	−0.21 [−0.30, −0.12]	−1.19 [−1.50, −0.96]	—
geometry	OPB-FS — CEB-energy	+0.32 [0.26, 0.40]	+2.67 [2.45, 2.92]	—
geometry (EPB)	EPB-FS — CEB-tied	—	—	+0.01 [−0.00, 0.01]
fixed scale	GPB — OPB-FS	−0.09 [−0.16, −0.02]	−0.04 [−0.07, −0.01]	—
fixed scale (EPB)	EPB — EPB-FS	—	—	+0.00 [−0.01, 0.01]
IB increment*	GPB — NCM-ortho	+0.34 [0.22, 0.45]	+1.89 [1.41, 2.36]	—
prototype structure	NCM-ortho — NCM-learn	−0.06 [−0.20, 0.11]	+0.16 [−0.49, 0.82]	—
prototype readout alone	NCM-learn — Base	−0.16 [−0.28, −0.05]	−0.96 [−1.30, −0.69]	—

Table 11: Factor-wise paired differences (accuracy points on MNIST and ImageNet-100, R^2 on Housing; 95% bootstrap percentile CIs, $B = 10,000$, fixed seed, paired across the shared five seeds at each variant’s best configuration; positive values favor the first term). *NCM-ortho fixes the frame at $a e_k$ while GPB trains it, so the IB comparison additionally contains frame trainability.

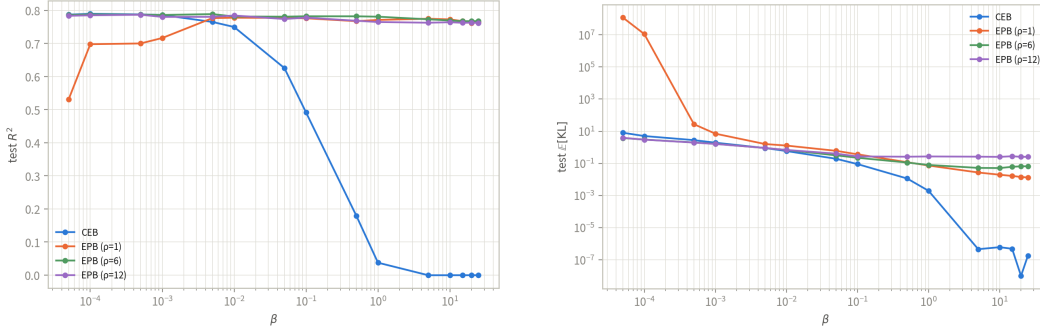


Figure 9: California Housing compression sweep (MLP). Left: test R^2 ; right: realized conditional KL over β . EPB maintains a task plateau through strong compression, while the $\rho = 1$ weak-compression points exhibit a large variance term.

B.4 INFORMATION-PLANE ESTIMATES

The information-plane quantities use the InfoNCE and cross-entropy estimators described in the former main-text analysis. They are estimator-level diagnostics, not information identities. On MNIST, CEB moves from $(I(X; Z), \text{Acc}) = (2.53, 98.2\%)$ at $\beta = 0.1$ to $(1.96, 86.0\%)$ at $\beta = 0.5$ and then collapses; OPB with $a = 6, 12$ retains 98.1–98.6% across the sweep. The MNIST panels appear in Fig. 4(c,d) of the main text; the matched-compression table follows.

B.5 ADVERSARIAL ROBUSTNESS DETAILS

Targeted PGD uses 20 steps, EOT-10 gradients, 10-sample MC predictions, and budgets $\varepsilon_\infty = 0.3$ and $\varepsilon_2 = 6$. Curves are shown only when clean stochastic accuracy exceeds 90%. At maximum compression, OPB ($a = 6, 12$) reaches 31.1%/30.5% targeted success under L_∞ , while transfer attacks from CEB to OPB reach at most 2.0%/3.4% under L_∞/L_2 , far below OPB’s white-box attack. The transfer result is a gradient-masking check rather than a causal geometry test.

C RELATED WORK

Information bottleneck and conditional compression. The information bottleneck (IB) principle formulates representation learning as retaining information about a target while discarding information about the input (Tishby et al., 2000). Deep VIB turns this principle into a neural objective by using a reparameterized stochastic encoder and a variational marginal prior (Alemi et al., 2017). Nonlinear and non-parametric bottleneck variants study alternative variational bounds and distance-based estimates of the information in a representation (Kolchinsky et al., 2019). The deterministic-scenario analysis of Kolchinsky & Tracey (2019) further shows that the usual IB multiplier can fail to sweep an interior information–accuracy trade-off when labels are deterministic; the squared-compression baseline in our experiments follows this line of work.

Setting	CEB $I(X; Z)$	CEB Acc.	OPB config.	OPB $I(X; Z)$	OPB Acc.
MNIST, $\beta = 0.1$	2.53	98.2	$a = 1, \beta = 0.1$	2.55	98.1
MNIST, $\beta = 0.5$	1.96	86.0	$a = 1, \beta = 0.5$	2.01	97.8

Table 12: Matched-compression information-plane comparison (means; standard deviations are in the stored evaluation report).

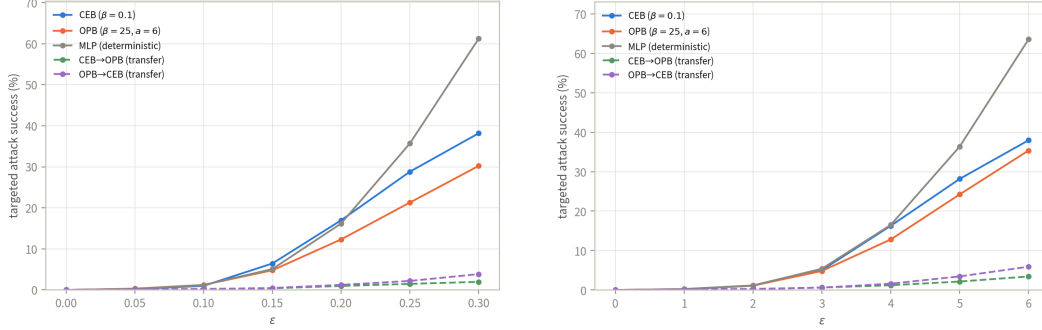


Figure 10: Transfer PGD curves on MNIST (gradient-masking check). Targeted adversaries are generated on one model and evaluated on another; CEB-to-OPB transfer remains far below OPB’s own white-box attack. The targeted white-box curves are in the main text.

The conditional entropy bottleneck (CEB) replaces the marginal reference of VIB by a label-conditional reference and thereby targets conditional residual information (Fischer, 2020). Comparisons of variational bounds clarify when this conditional reference is tighter than a marginal one (Geiger & Fischer, 2020). CEB has also been applied to large-scale visual representations and robustness evaluations (Fischer & Alemi, 2020; Lee et al., 2021), while multivariate extensions retain the same variational bottleneck viewpoint (Abdelaleem et al., 2025). These methods leave the relationships among distinct conditional reference means unspecified. GPB keeps the CEB posterior and its single conditional KL, but restricts the reference family itself: OPB declares a class frame and EPB declares a label-distance-preserving axis.

Information bottleneck for robust representations. Several works use a bottleneck to suppress nuisance or non-robust information. The HSIC bottleneck adds layer-wise dependence penalties and analyzes adversarial robustness (Wang et al., 2021). Adversarial Information Bottleneck explicitly incorporates adversarial objectives (Zhai & Zhang, 2022), and an NLP-specific variant separates task-relevant robust features from manipulable features (Zhang et al., 2022). Causal Information Bottleneck uses a causal decomposition for the same robust-feature goal (Hua et al., 2022). These approaches modify the information criterion, feature selection, or training examples. In contrast, the geometric constraint in GPB is placed on the conditional prior and is evaluated with the ordinary CEB KL; the robustness experiments therefore test whether a declared prior geometry is transferred to the posterior rather than introducing adversarial training as an additional objective.

Geometric classification representations. Neural-collapse studies document an approximately simplex-ETF arrangement of class means and classifier directions near the terminal phase of training (Papayan et al., 2020; Zhu et al., 2021). The connection between neural collapse, supervised contrastive learning, and an information bottleneck has also been analyzed explicitly (Wang & Palmer, 2023). Other works impose related geometry directly: orthogonal classifier weights can improve adversarial robustness (Xu et al., 2022); ETF classifiers have been used for early exit (Ji et al., 2023); and fixed or hierarchy-aware frames have been used to reduce classifier bias or mistake severity (Tong & Davis, 2023). Neural Collapse by Design trains features and prototypes jointly on the unit hypersphere so that the global optimum of the NONL contrastive objective is exactly the simplex ETF (Koromilas et al., 2026); we include it as a classification-only comparison baseline. Equidistant prototype embeddings provide an earlier prototype-based classification example (Deng et al., 2014).

These works constrain features, classifier weights, or prototypes through a classification loss or a fixed readout. OPB addresses a different object: the label-conditional Gaussian prior in the bottle-

Approach	Constrained object	Enters an IB	Readout tied to	Stochastic posterior?
Prototype methods (Deng et al., 2014)	prototypes or features	No	Usually no/partial	Usually no
ETF / orthogonal classifiers (Xu et al., 2022; Ji et al., 2023)	classifier weights or feature directions	No	Yes	Usually no
Fixed or hierarchy-aware frames (Tong & Davis, 2023)	feature coordinates or a frame	No	Method-dependent	Usually no
CEB (Fischer, 2020)	label-conditional reference	Yes (conditional KL)	No (free head)	Yes
GPB (OPB / EPB)	label-conditional Gaussian prior means and covariance	Yes (conditional KL)	Yes (tied rule)	Yes

Table 13: Conceptual boundary between geometric representation methods and GPB. The entries describe the object constrained by the method, not whether a particular implementation may contain additional components. GPB reuses familiar geometric primitives while placing them in the conditional prior used by the bottleneck and using the same prior geometry for prediction.

neck. Its polar/QR frame is recomputed from the prior encoder, and the anchor-energy classifier reads the same frame without introducing an independent classifier scale. Thus the geometric class code and the conditional KL are coupled during training.

Table 13 summarizes this object-level distinction. The geometric primitive is shared with several preceding lines of work, but its role in the objective is different: OPB anchors are constrained class centers that serve simultaneously as Gaussian-prior means, KL targets, and readout anchors. In the shared-covariance canonical instance, the induced readout is an orthogonal classifier as a consequence of the Gaussian Bayes rule. It is not an ETF construction: OPB has Gram matrix $a^2 I_K$ (zero pairwise inner products), whereas a centered simplex ETF has constant negative off-diagonal inner products in a $(K - 1)$ -dimensional subspace. The frame is recomputed from the prior encoder and may rotate, rather than fixing a coordinate frame or optimizing a separate prototype table.

Distance-preserving regression and geometric priors. Isometric representation learning has traditionally focused on preserving distances on a data manifold, including deep extensions of isomap-style mappings (Pai et al., 2022). Orthogonal label regression has also been used as a discriminative device in domain adaptation (Luo et al., 2023). These methods motivate using a declared label metric, but they generally constrain a deterministic feature map rather than the mean map of a conditional Gaussian information-bottleneck prior. EPB uses the scalar label case: after standardization, prior means lie on a normalized axis with a fixed scale, so latent prior distances equal the declared label distances up to that scale. The same axis is also the tied regression readout and the target of the conditional KL. Thus EPB is a conditional-prior use of distance geometry, not a new distance-preserving primitive. The regression information-bottleneck literature, including divergence-based bounds, provides complementary compression objectives rather than this geometric prior construction (Yu et al., 2024). AdaCap pairs a permutation-based contrastive loss with a Tikhonov closed-form output layer for small-sample tabular regression (Belucci et al., 2026); we include it as a regression-only comparison baseline.

Geometry-aware information bottlenecks. Recent work has used the phrase geometric information bottleneck in several distinct settings. A variational recommender learns a geometric prior for explanation generation (Yan et al., 2024); a data-scarcity formulation penalizes latent curvature and intrinsic dimension (Katende, 2025); and activation steering uses orthogonal subspaces to control language-model interventions (Doan et al., 2026). A concurrent analysis studies the relationship between information compression and class-wise geometric compression in CEB networks (Adilova

et al., 2026). These studies make geometry an important representation concern, but they do not use the same conditional Gaussian prior for both compression and prediction, nor do they provide the OPB/EPB alignment-to- separation construction considered here.